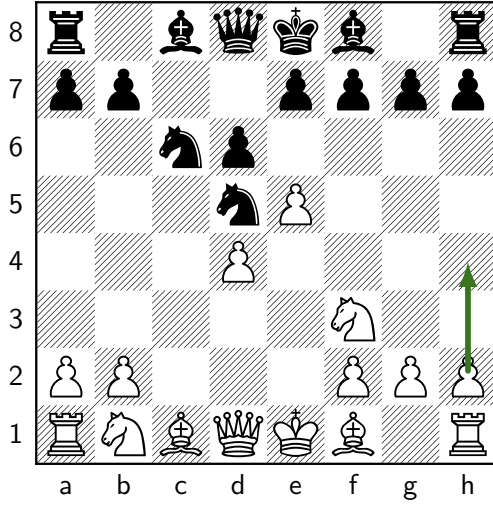


Melanie-BOTS (CCRL engines) — Sicilian, Alapin

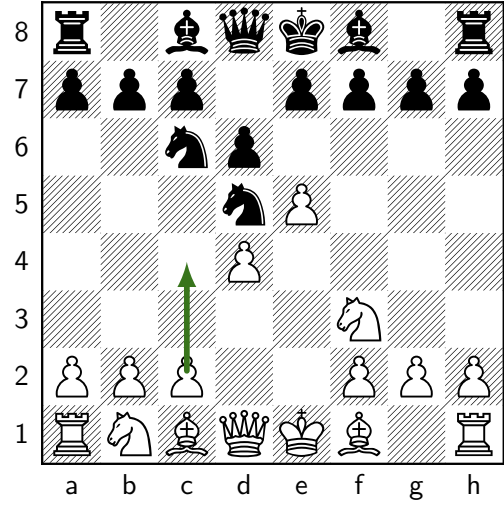
Melanie (Deep_Vision)

Abstract

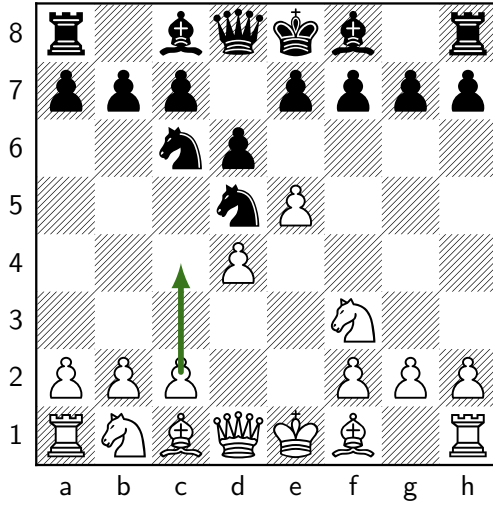
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



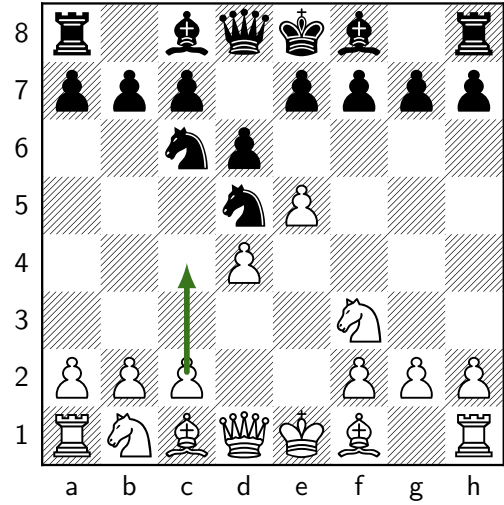
(a)



(b)

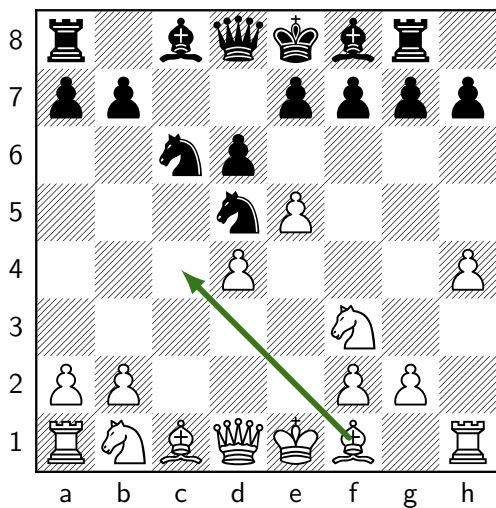


(c)

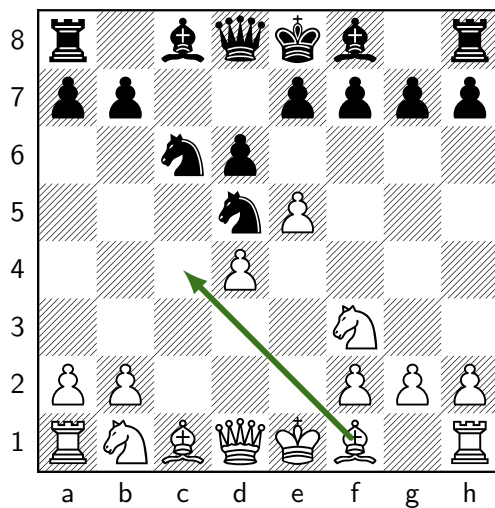


(d)

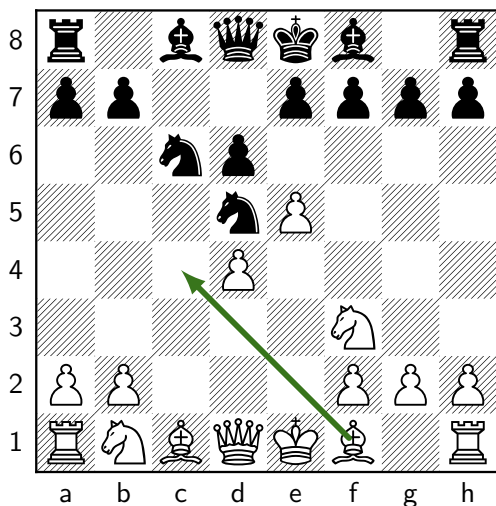
Figure 1: **Move 7, White: h4.** (a) Our position; Melanie plays **h4** (**h2h4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2888` (ply 8) [structural distance $d=33$, QAP obj=128]: the master played **c4**; under μ our *Ph2* plays the role of that game’s *Pc2*; cross-type roles: our *Bc1* as their *Nb1*, our *Qd1* as their *Bc1*; transferred move **h2h4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2889` (ply 8) [structural distance $d=33$, QAP obj=128]: the master played **c4**; under μ our *Ph2* plays the role of that game’s *Pc2*; cross-type roles: our *Bc1* as their *Nb1*, our *Qd1* as their *Bc1*; transferred move **h2h4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1779` (ply 8) [structural distance $d=33$, QAP obj=128]: the master played **c4**; under μ our *Ph2* plays the role of that game’s *Pc2*; cross-type roles: our *Bc1* as their *Nb1*, our *Qd1* as their *Bc1*; transferred move **h2h4** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **h4** (summed QAP objective 384, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



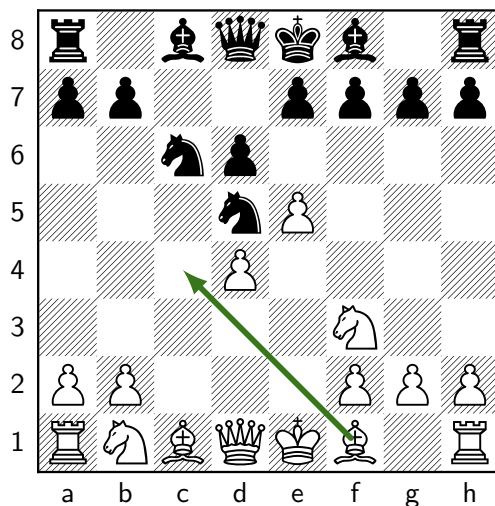
(a)



(b)

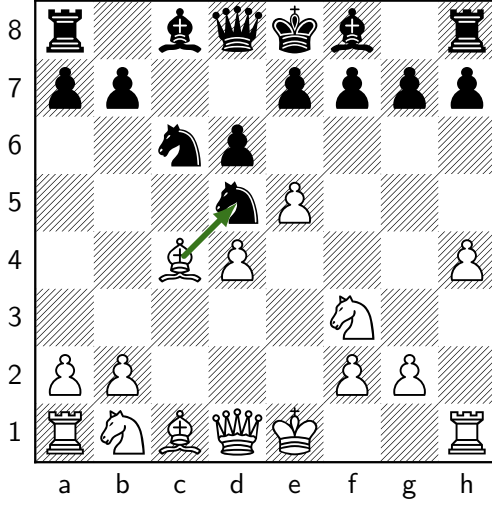


(c)

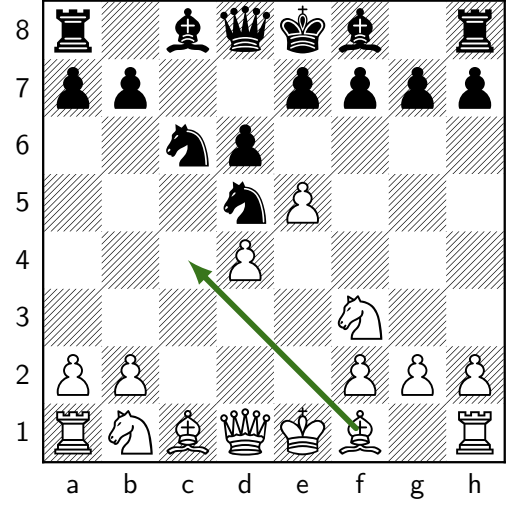


(d)

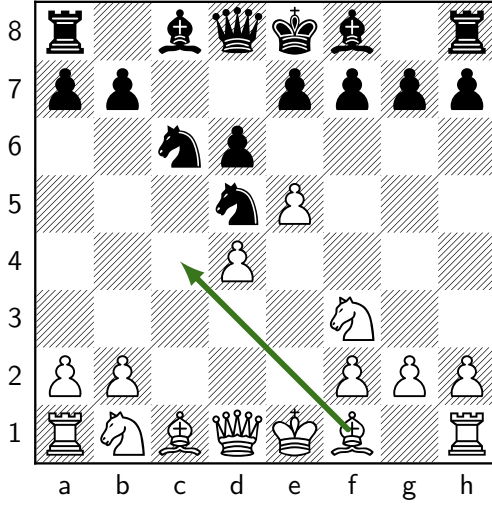
Figure 2: **Move 8, White: Bc4.** (a) Our position; Melanie plays Bc4 (f1c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#534` (ply 12) [structural distance $d=1$, QAP obj=132]: the master played Bc4; under μ our Bf1 plays the role of that game’s Bf1; cross-type roles: our Pg2 as their Nb1, our Nb1 as their Pa2; transferred move f1c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#535` (ply 12) [structural distance $d=1$, QAP obj=132]: the master played Bc4; under μ our Bf1 plays the role of that game’s Bf1; cross-type roles: our Pg2 as their Nb1, our Nb1 as their Pa2; transferred move f1c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2752` (ply 12) [structural distance $d=1$, QAP obj=132]: the master played Bc4; under μ our Bf1 plays the role of that game’s Bf1; cross-type roles: our Pg2 as their Nb1, our Nb1 as their Pa2; transferred move f1c4 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to Bc4 (summed QAP objective 1712, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



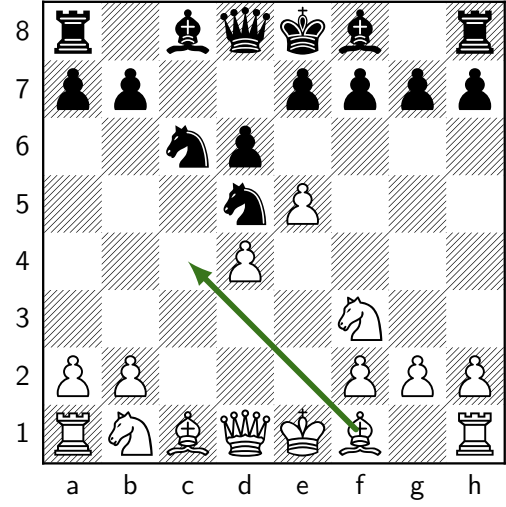
(a)



(b)

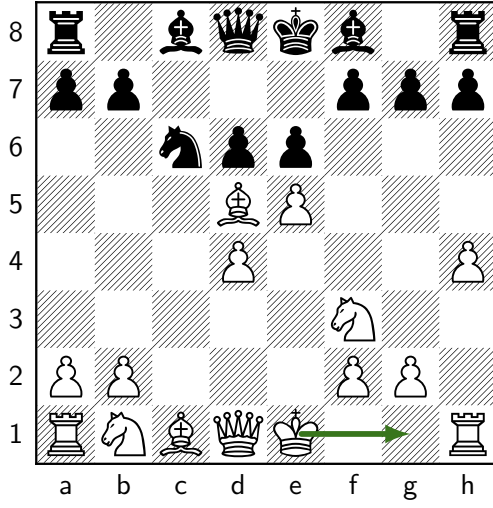


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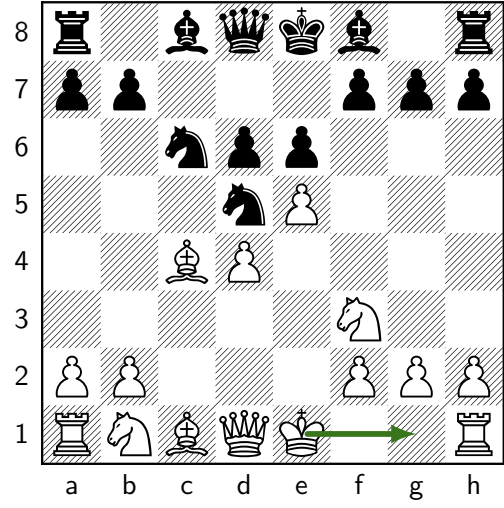


(d)

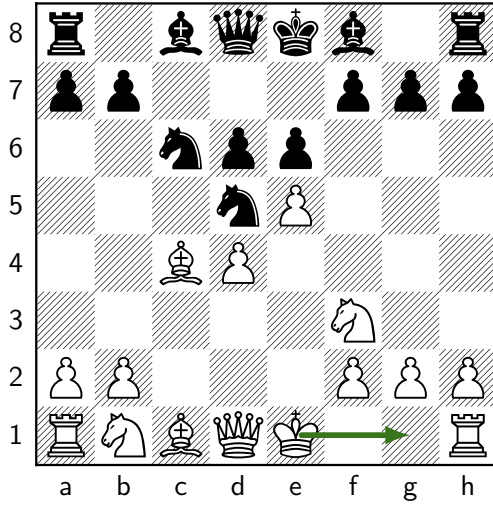
Figure 3: **Move 9, White: Bxd5.** (a) Our position; Melanie plays Bxd5 (c4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#534` (ply 12) [structural distance $d=16$, QAP obj=75]: the master played Bc4; under μ our Bc4 plays the role of that game’s Bf1; cross-type roles: our bc8 as their Ra1, our Ra1 as their Nb1; transferred move c4b5. (c) Retrieved from `ccr1_top_p0000#535` (ply 12) [structural distance $d=16$, QAP obj=75]: the master played Bc4; under μ our Bc4 plays the role of that game’s Bf1; cross-type roles: our bc8 as their Ra1, our Ra1 as their Nb1; transferred move c4b5. (d) Retrieved from `ccr1_top_p0000#2752` (ply 12) [structural distance $d=16$, QAP obj=75]: the master played Bc4; under μ our Bc4 plays the role of that game’s Bf1; cross-type roles: our bc8 as their Ra1, our Ra1 as their Nb1; transferred move c4b5. Synthesis: no retrieved analog transferred to Bxd5 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



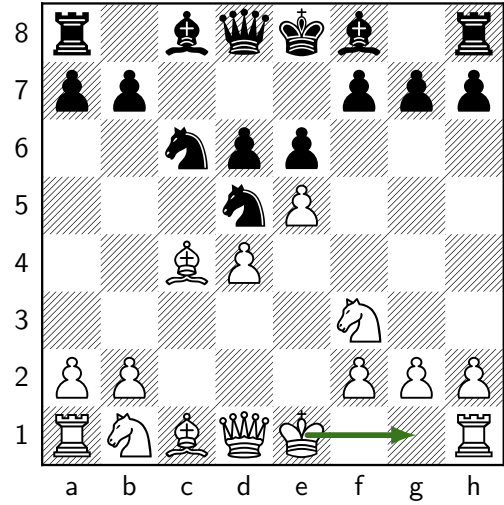
(a)



(b)

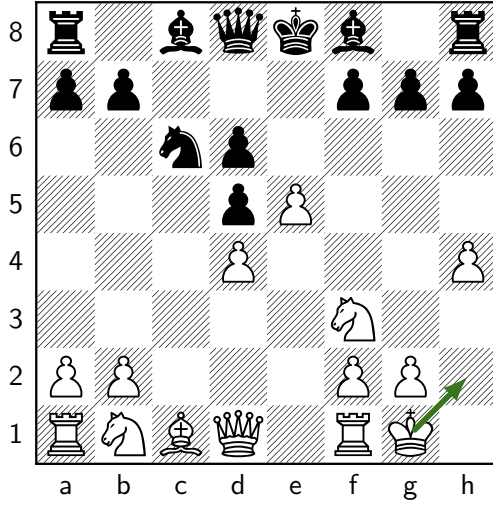


(c)

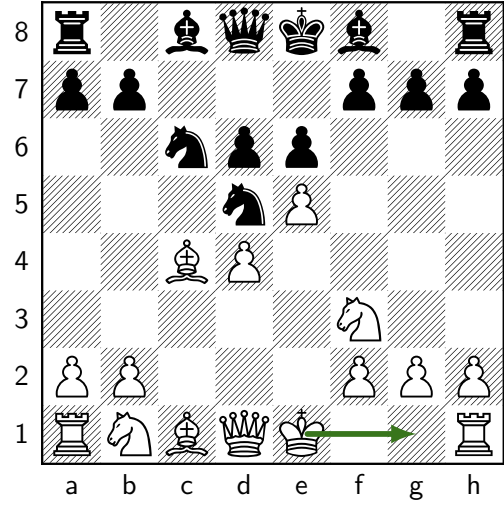


(d)

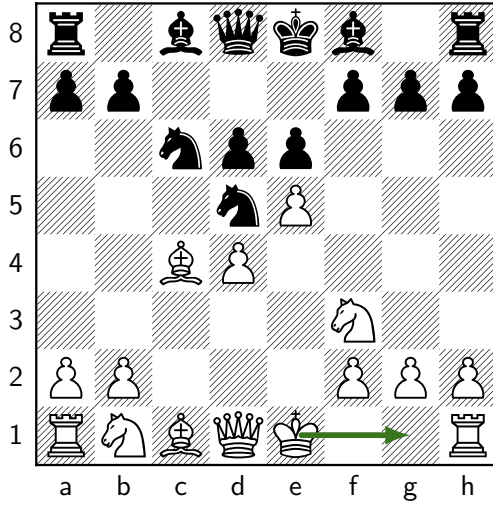
Figure 4: **Move 10, White: 0-0.** (a) Our position; Melanie plays 0-0 (**e1g1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2205` (ply 14) [structural distance $d=37$, QAP obj=58]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *pg7* as their *Ra1*, our *Pg2* as their *Nb1*; transferred move **e1g1** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2206` (ply 14) [structural distance $d=37$, QAP obj=58]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *pg7* as their *Ra1*, our *Pg2* as their *Nb1*; transferred move **e1g1** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2207` (ply 14) [structural distance $d=37$, QAP obj=58]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *pg7* as their *Ra1*, our *Pg2* as their *Nb1*; transferred move **e1g1** (**endorses this move**). Synthesis: 12 retrieved analogs transferred to 0-0 (summed QAP objective 701, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



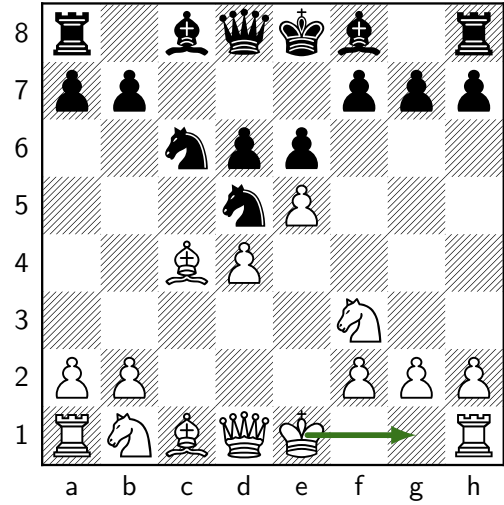
(a)



(b)

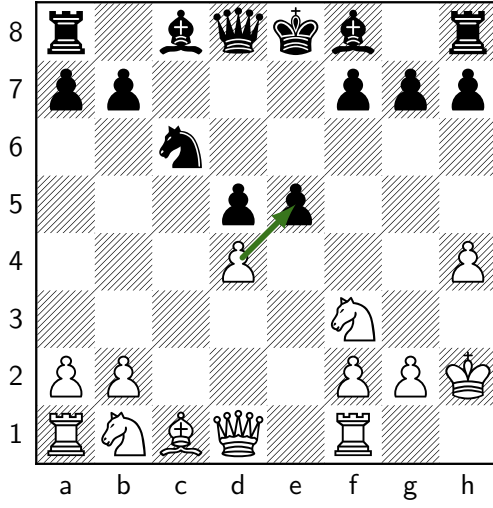


(c)

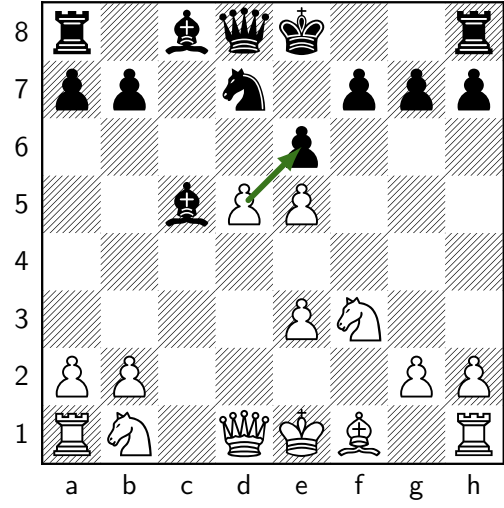


(d)

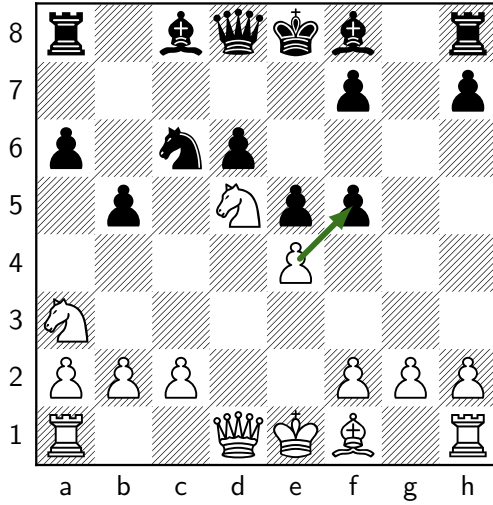
Figure 5: **Move 11, White: Kh2**. (a) Our position; Melanie plays Kh2 (**g1h2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2205` (ply 14) [structural distance $d=60$, QAP obj=56]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *bf8* as their *Ra1*, our *Pg2* as their *Nb1*; transferred move **g1h2 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#2206` (ply 14) [structural distance $d=60$, QAP obj=56]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *bf8* as their *Ra1*, our *Pg2* as their *Nb1*; transferred move **g1h2 (endorses this move)**. (d) Retrieved from `ccr1_top_p0000#2207` (ply 14) [structural distance $d=60$, QAP obj=56]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *bf8* as their *Ra1*, our *Pg2* as their *Nb1*; transferred move **g1h2 (endorses this move)**. Synthesis: 12 retrieved analogs transferred to Kh2 (summed QAP objective 673, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



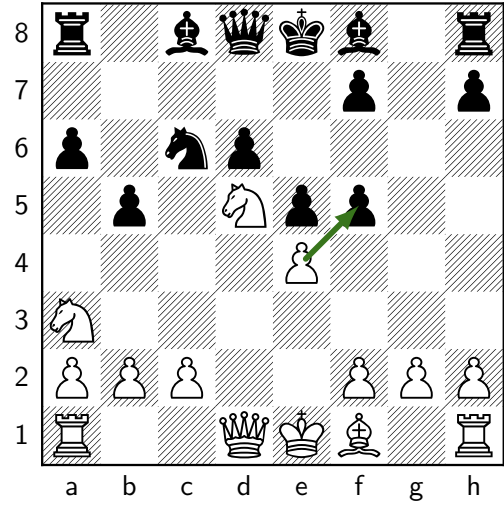
(a)



(b)

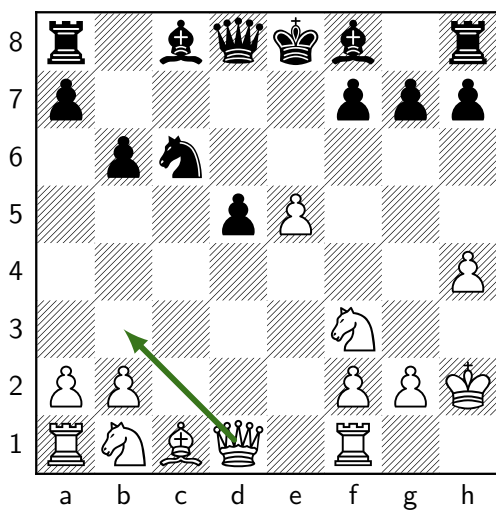


(c)

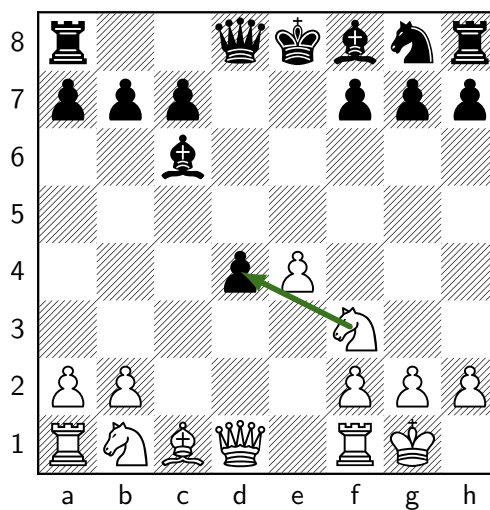


(d)

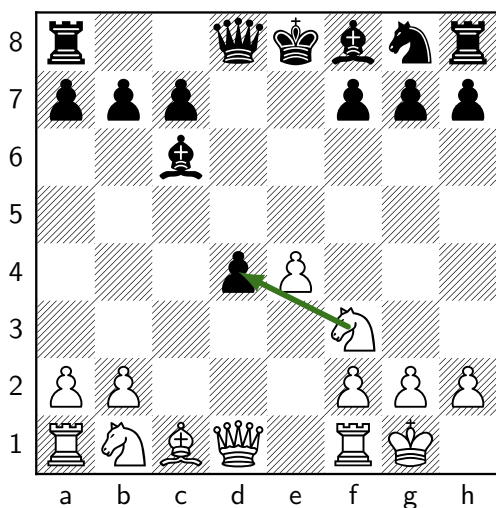
Figure 6: **Move 12, White: dxe5.** (a) Our position; Melanie plays dxe5 (d4e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#728 (ply 18) [structural distance $d=67$, QAP obj=53]: the master played dxe6; under μ our Pd4 plays the role of that game’s Pd5; cross-type roles: our Pf2 as their Ra1, our Pg2 as their Nb1; transferred move d4e5 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#212 (ply 22) [structural distance $d=71$, QAP obj=48]: the master played exf5; under μ our Pd4 plays the role of that game’s Pe4; cross-type roles: our Nf3 as their Bf1, our bf8 as their Rh1; transferred move d4e5 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#1646 (ply 22) [structural distance $d=71$, QAP obj=48]: the master played exf5; under μ our Pd4 plays the role of that game’s Pe4; cross-type roles: our Nf3 as their Bf1, our bf8 as their Rh1; transferred move d4e5 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to dxe5 (summed QAP objective 284, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



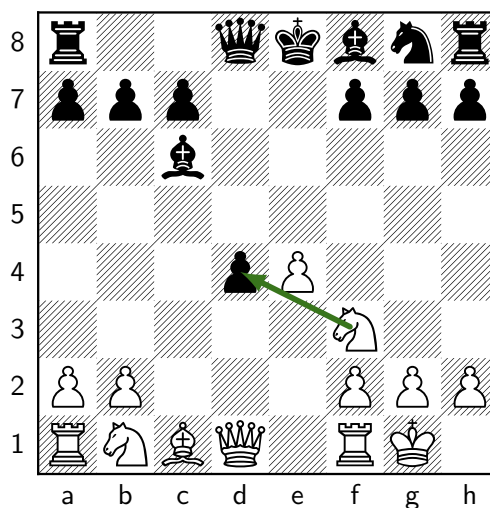
(a)



(b)

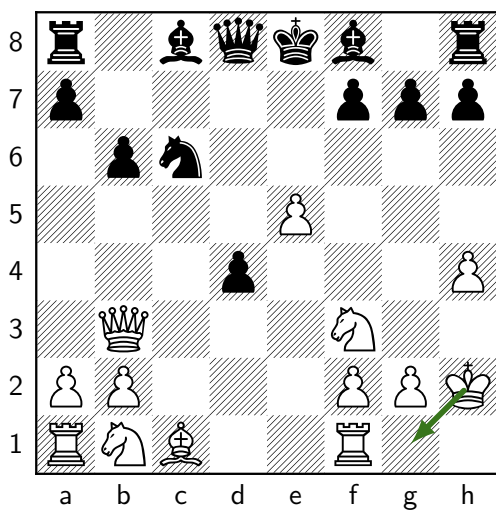


(c)

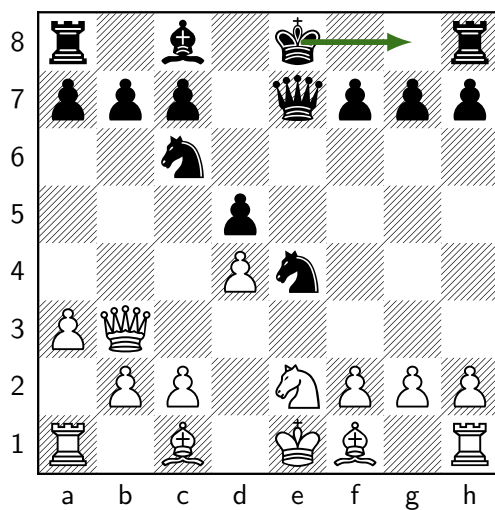


(d)

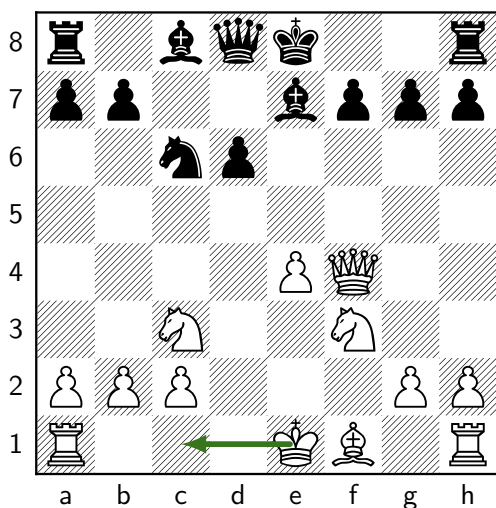
Figure 7: **Move 13, White: Qb3.** (a) Our position; Melanie plays Qb3 (d1b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1729` (ply 16) [structural distance $d=75$, QAP obj=55]: the master played Nxd4; under μ our Qd1 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Bc1 as their Ra1, our ph7 as their Nb1; transferred move d1b3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1730` (ply 16) [structural distance $d=75$, QAP obj=55]: the master played Nxd4; under μ our Qd1 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Bc1 as their Ra1, our ph7 as their Nb1; transferred move d1b3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2534` (ply 16) [structural distance $d=75$, QAP obj=55]: the master played Nxd4; under μ our Qd1 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Bc1 as their Ra1, our ph7 as their Nb1; transferred move d1b3 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Qb3 (summed QAP objective 328, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



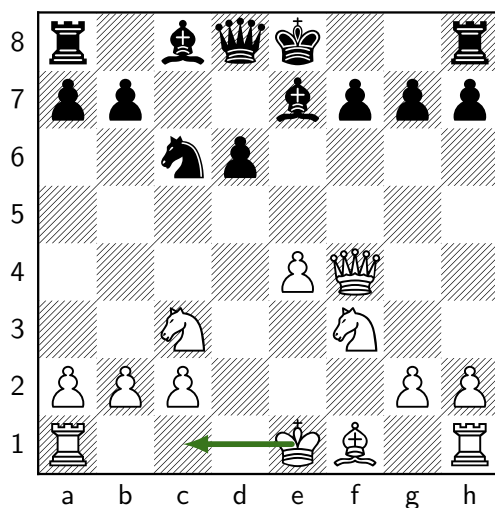
(a)



(b)

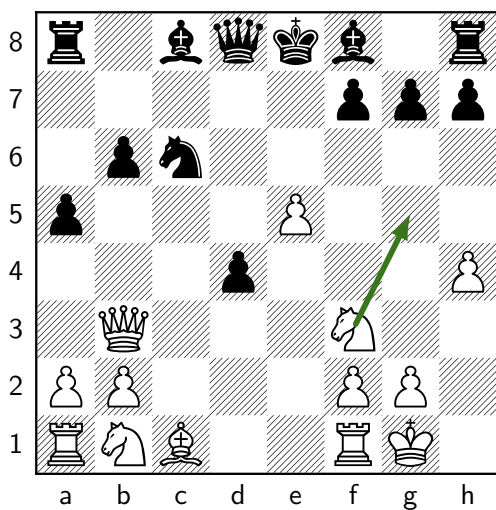


(c)

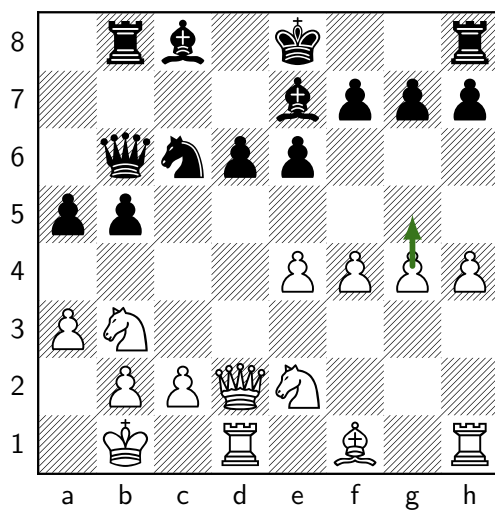


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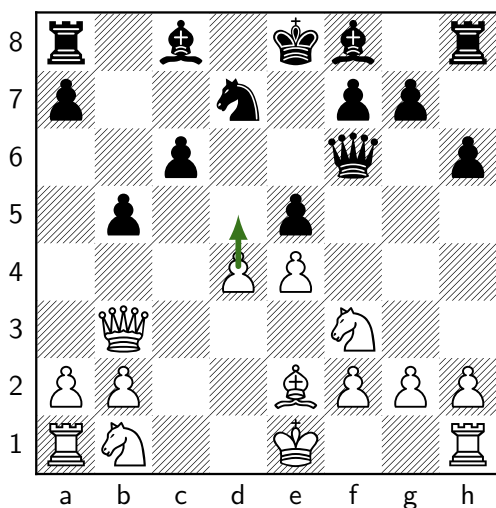
Figure 8: **Move 14, White: Kg1**. (a) Our position; Melanie plays Kg1 (h2g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1468` (ply 19) [structural distance $d=78$, QAP obj=65]: the master played 0-0; under μ our *Kh2* plays the role of that game’s *Ke1*; cross-type roles: our *Nb1* as their *Rh8*, our *Bc1* as their *Pa7*; transferred move h2g1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#152` (ply 22) [structural distance $d=71$, QAP obj=33]: the master played 0-0-0; under μ our *Kh2* plays the role of that game’s *Ke1*; cross-type roles: our *Bc1* as their *Ra1*, our *Ra1* as their *Pc2*; transferred move h2g1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#949` (ply 22) [structural distance $d=71$, QAP obj=33]: the master played 0-0-0; under μ our *Kh2* plays the role of that game’s *Ke1*; cross-type roles: our *Bc1* as their *Ra1*, our *Ra1* as their *Pc2*; transferred move h2g1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kg1 (summed QAP objective 134, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



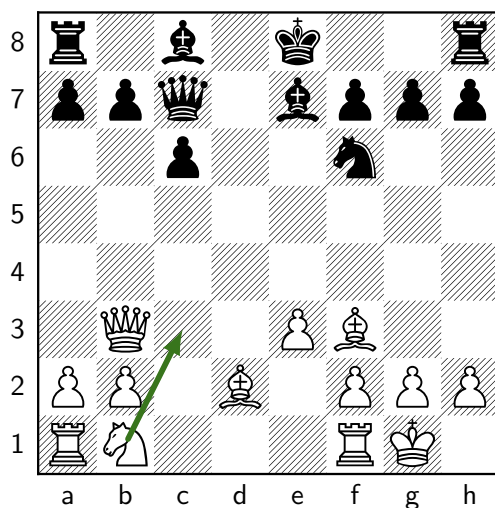
(a)



(b)

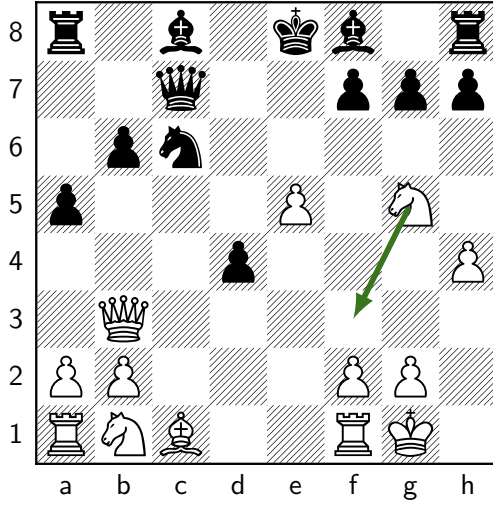


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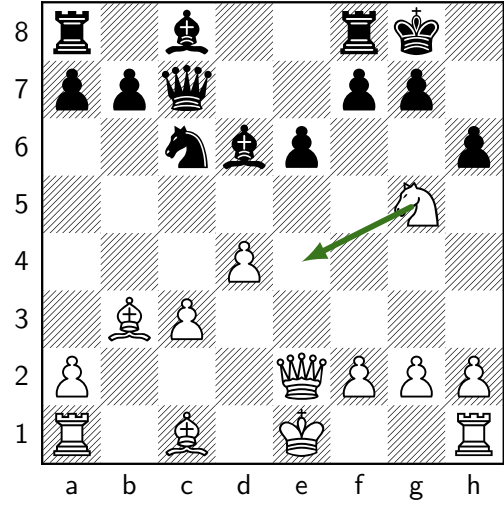


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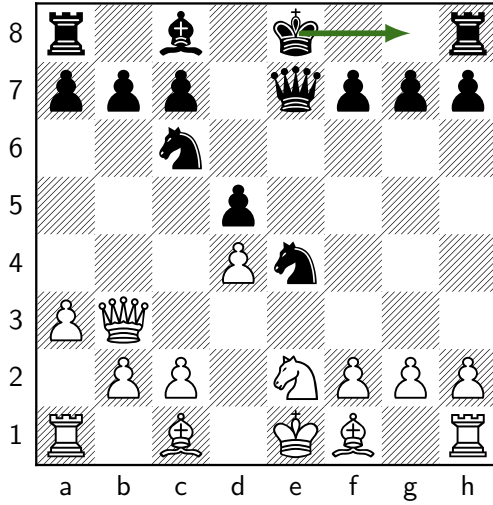
Figure 9: **Move 15, White: Ng5.** (a) Our position; Melanie plays Ng5 (f3g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3906` (ply 32) [structural distance $d=75$, QAP obj=48]: the master played g5; under μ our *Nf3* plays the role of that game’s *Pg4* (a **cross-type** correspondence); cross-type roles: our *Pb2* as their *Bf1*, our *ph7* as their *Rh1*; transferred move f3g5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2177` (ply 20) [structural distance $d=74$, QAP obj=35]: the master played d5; under μ our *Nf3* plays the role of that game’s *Pd4* (a **cross-type** correspondence); cross-type roles: our *Bc1* as their *Ra1*, our *Ph4* as their *Nb1*; transferred move f3g5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1433` (ply 22) [structural distance $d=76$, QAP obj=49]: the master played Nc3; under μ our *Nb1* plays the role of that game’s *Nb1*; cross-type roles: our *bf8* as their *Ra1*, our *rh8* as their *Pb2*; transferred move b1c3. Synthesis: 2 retrieved analogs transferred to Ng5 (summed QAP objective 83, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



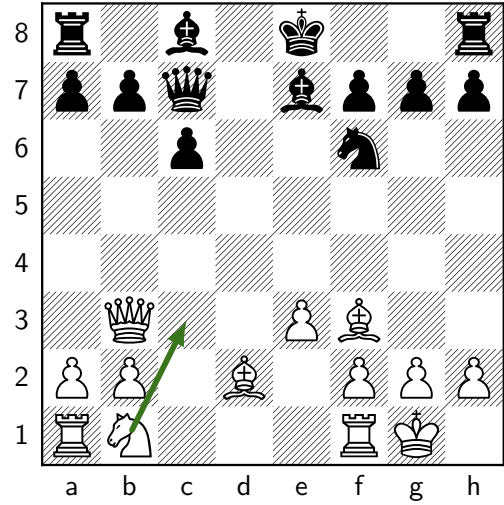
(a)



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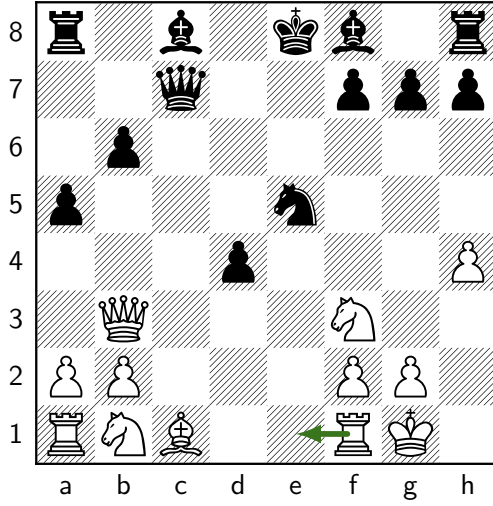


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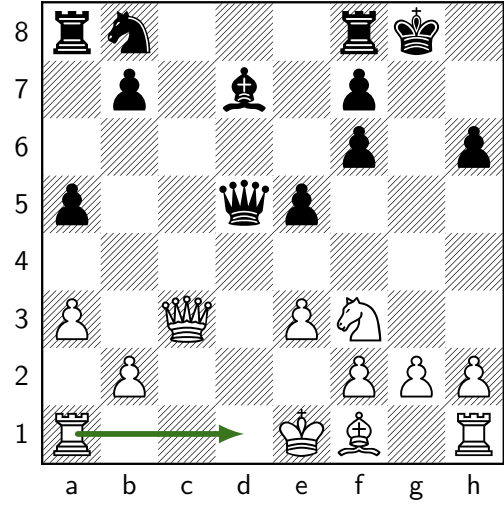


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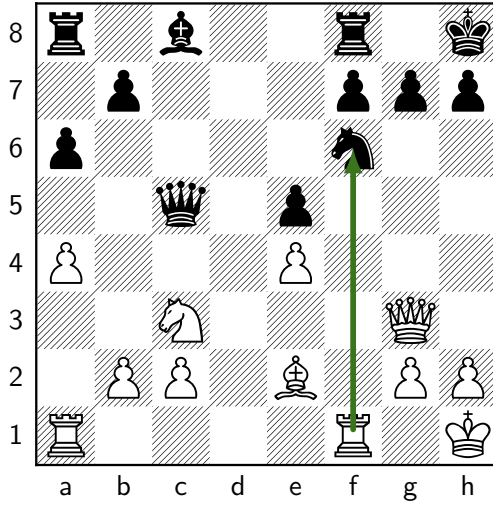
Figure 10: **Move 16, White: Nf3.** (a) Our position; Melanie plays Nf3 (g5f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#479` (ply 24) [structural distance $d=88$, QAP obj=6]: the master played Ne4; under μ our Ng5 plays the role of that game’s Ng5; cross-type roles: our Bc1 as their Ra1, our Ph4 as their Bc1; transferred move g5f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1468` (ply 19) [structural distance $d=90$, QAP obj=77]: the master played 0-0; under μ our Kg1 plays the role of that game’s Ke1; cross-type roles: our Nb1 as their Ra8, our ra8 as their Bc8; transferred move g1h1. (d) Retrieved from `ccr1_top_p0003#1433` (ply 22) [structural distance $d=87$, QAP obj=49]: the master played Nc3; under μ our Nb1 plays the role of that game’s Nb1; cross-type roles: our nc6 as their Ra1, our rh8 as their Pb2; transferred move b1c3. Synthesis: 1 retrieved analog transferred to Nf3 (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



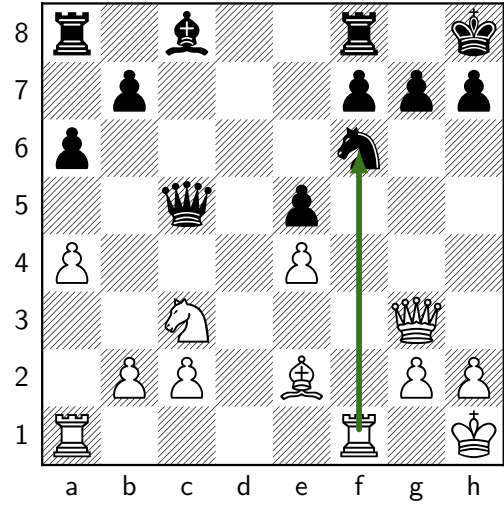
(a)



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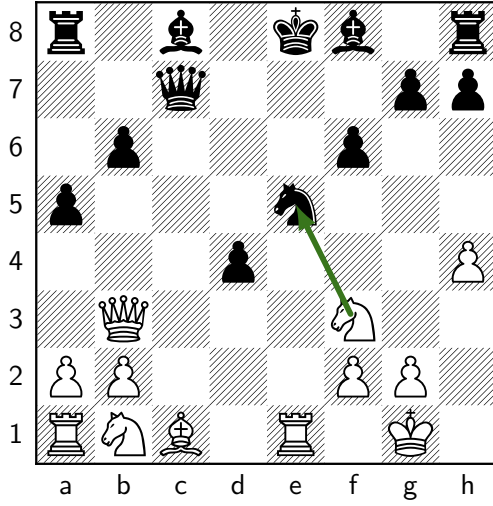


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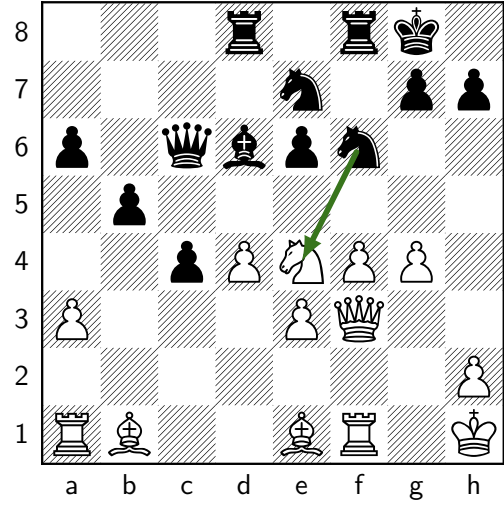


(d)

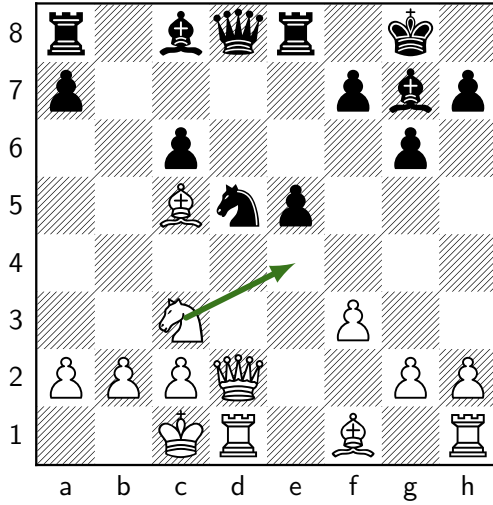
Figure 11: **Move 17, White: Re1.** (a) Our position; Melanie plays Re1 (f1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#380` (ply 28) [structural distance $d=83$, QAP obj=42]: the master played Rd1; under μ our Rf1 plays the role of that game’s Ra1; cross-type roles: our Qb3 as their Bf1, our Nb1 as their Pg2; transferred move f1e1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2472` (ply 32) [structural distance $d=68$, QAP obj=34]: the master played Rxf6; under μ our Rf1 plays the role of that game’s Rf1; cross-type roles: our ph7 as their Ra1, our ra8 as their Be2; transferred move f1e1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#244` (ply 36) [structural distance $d=68$, QAP obj=34]: the master played Rxf6; under μ our Rf1 plays the role of that game’s Rf1; cross-type roles: our ph7 as their Ra1, our ra8 as their Be2; transferred move f1e1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Re1 (summed QAP objective 110, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



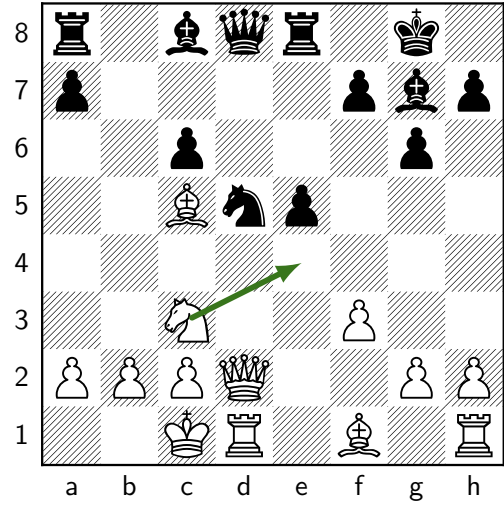
(a)



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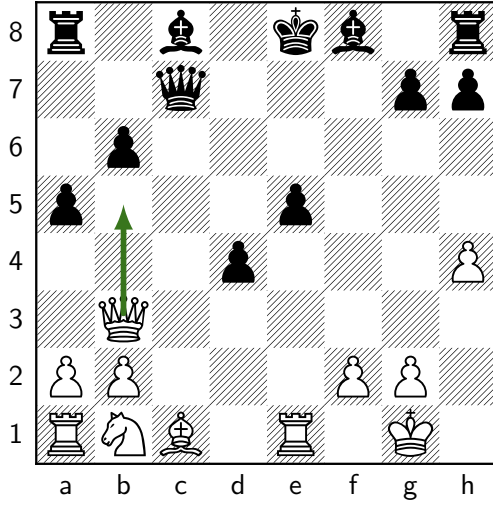


(c)

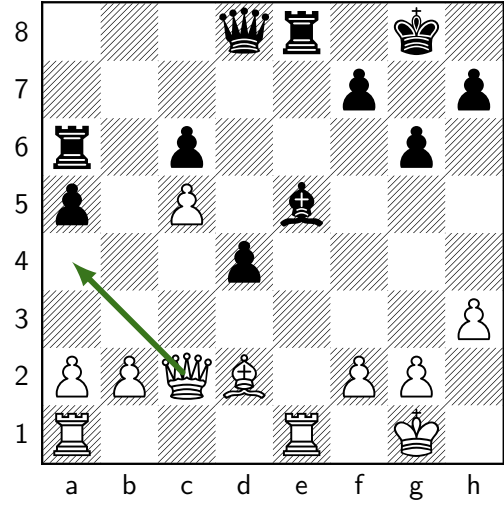


(d)

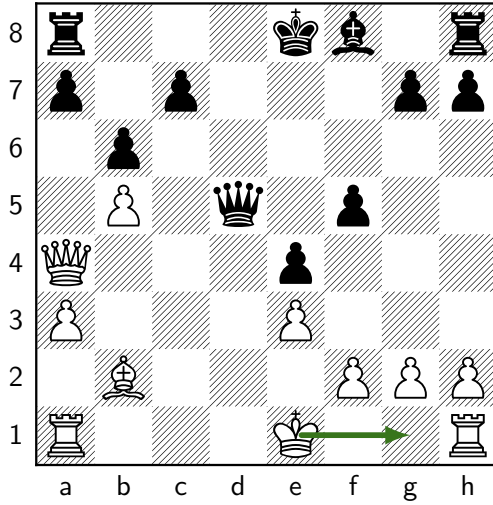
Figure 12: **Move 18, White: Nxe5.** (a) Our position; Melanie plays Nxe5 (f3e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2787` (ply 49) [structural distance $d=93$, QAP obj=132]: the master played Nxe4; under μ our Nf3 plays the role of that game's Nf3; cross-type roles: our ph7 as their Rd8, our Bc1 as their Ne7; transferred move f3e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2349` (ply 26) [structural distance $d=88$, QAP obj=96]: the master played Ne4; under μ our Nf3 plays the role of that game's Nc3; cross-type roles: our ph7 as their Rd1, our Pg2 as their Bf1; transferred move f3d4. (d) Retrieved from `ccr1_top_p0003#1594` (ply 26) [structural distance $d=88$, QAP obj=96]: the master played Ne4; under μ our Nf3 plays the role of that game's Nc3; cross-type roles: our ph7 as their Rd1, our Pg2 as their Bf1; transferred move f3d4. Synthesis: 1 retrieved analog transferred to Nxe5 (summed QAP objective 132, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



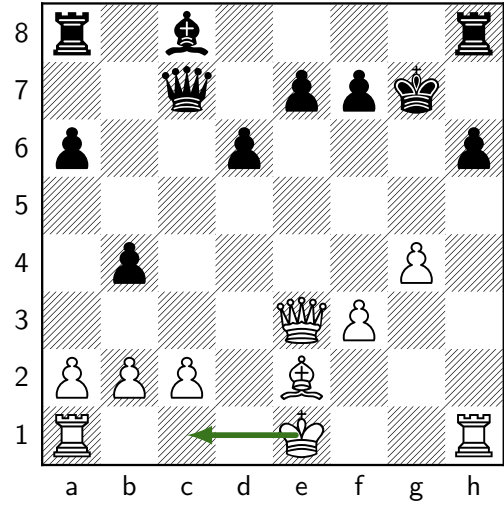
(a)



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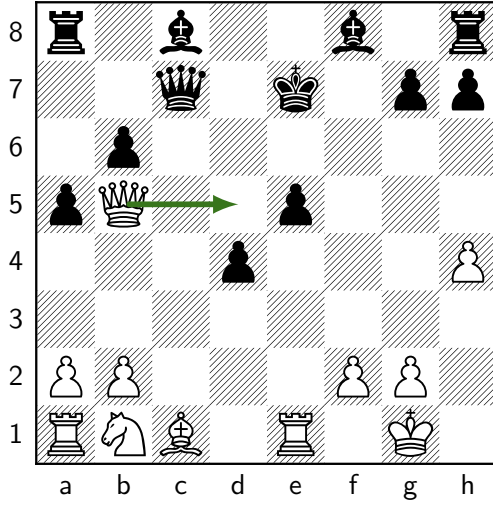


(c)

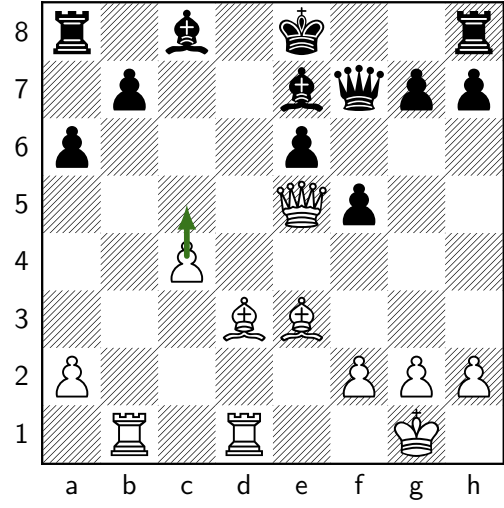


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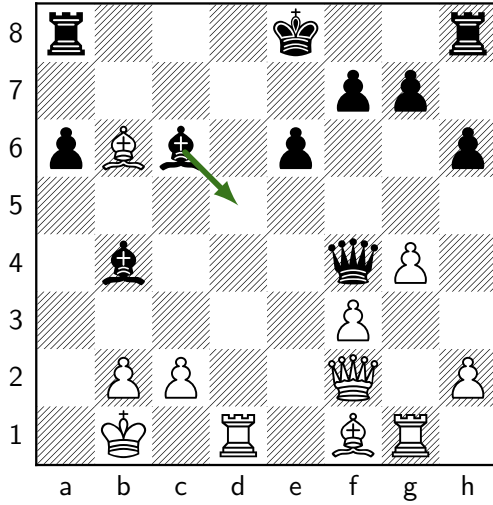
Figure 13: **Move 19, White: Qb5+.** (a) Our position; Melanie plays Qb5+ (b3b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#369` (ply 40) [structural distance $d=78$, QAP obj=34]: the master played Qa4; under μ our Qb3 plays the role of that game’s Qc2; cross-type roles: our Nb1 as their Ra1, our qc7 as their Bd2; transferred move b3b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2471` (ply 36) [structural distance $d=72$, QAP obj=59]: the master played 0-0; under μ our Kg1 plays the role of that game’s Ke1; cross-type roles: our Pb2 as their Ra1, our Qb3 as their Bb2; transferred move g1f1. (d) Retrieved from `ccr1_top_p0000#204` (ply 38) [structural distance $d=74$, QAP obj=56]: the master played 0-0-0; under μ our Kg1 plays the role of that game’s Ke1; cross-type roles: our Pg2 as their Ra1, our Ra1 as their Pa2; transferred move g1f1. Synthesis: 1 retrieved analog transferred to Qb5+ (summed QAP objective 34, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



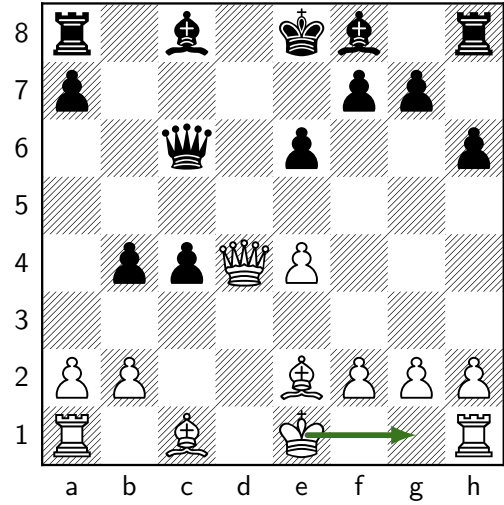
(a)



(b)

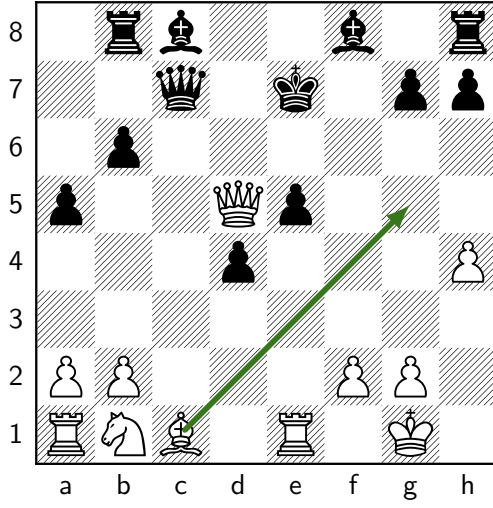


(c)

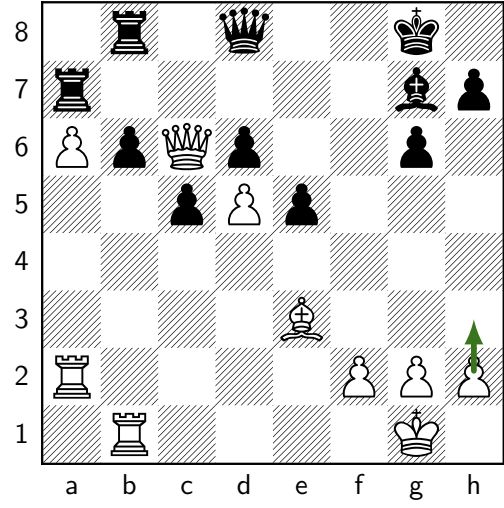


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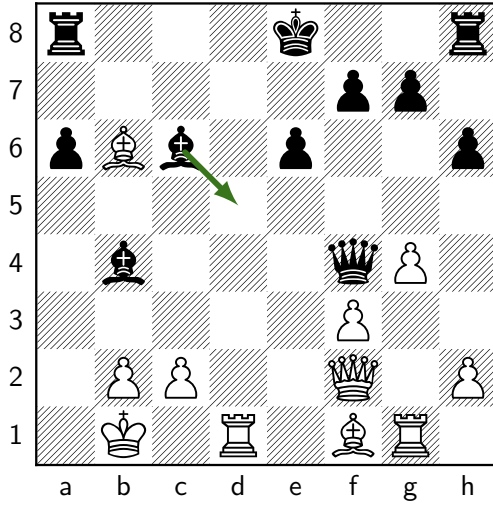
Figure 14: **Move 20, White: Qd5.** (a) Our position; Melanie plays Qd5 (b5d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#944` (ply 40) [structural distance $d=78$, QAP obj=60]: the master played c5; under μ our Pf2 plays the role of that game’s Pc4; cross-type roles: our Bc1 as their Rb1, our Ra1 as their Pa2; transferred move f2f4. (c) Retrieved from `ccr1_top_p0002#3741` (ply 37) [structural distance $d=79$, QAP obj=58]: the master played Bd5; under μ our Qb5 plays the role of that game’s Bc3 (a **cross-type** correspondence); cross-type roles: our Nb1 as their Rh8, our Bc1 as their Pg7; transferred move b5c4. (d) Retrieved from `ccr1_top_p0000#942` (ply 28) [structural distance $d=86$, QAP obj=53]: the master played 0-0; under μ our Kg1 plays the role of that game’s Ke1; cross-type roles: our Ph4 as their Bc1, our Pf2 as their Rh1; transferred move g1f1. Synthesis: no retrieved analog transferred to Qd5 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



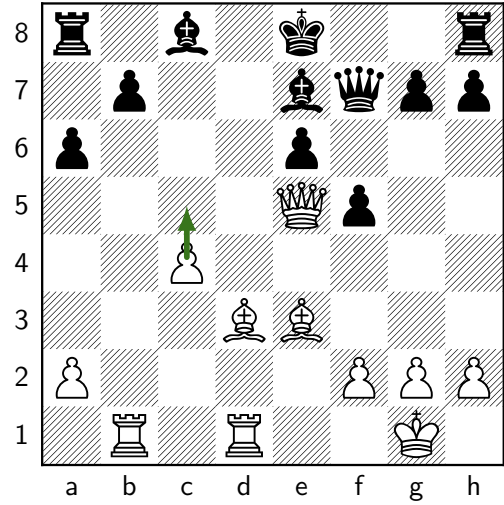
(a)



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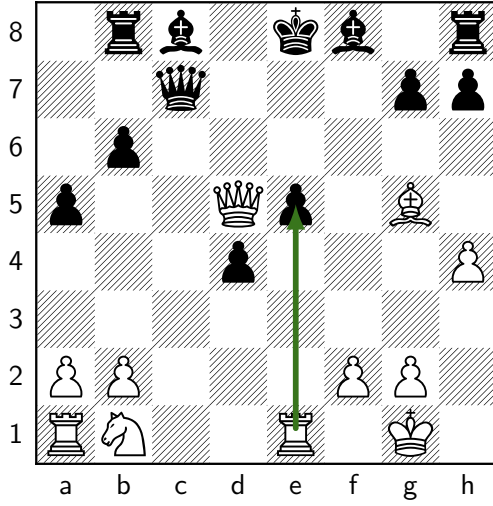


(c)

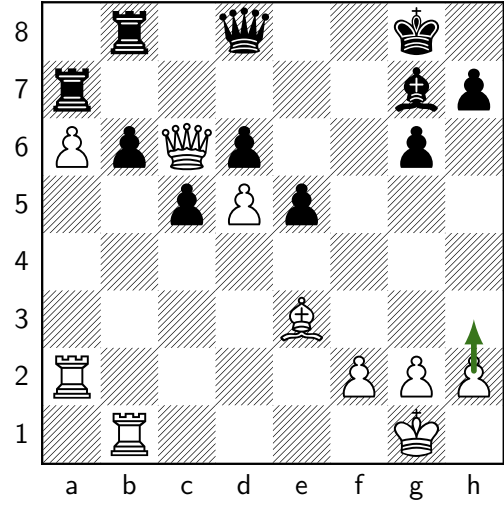


(d)

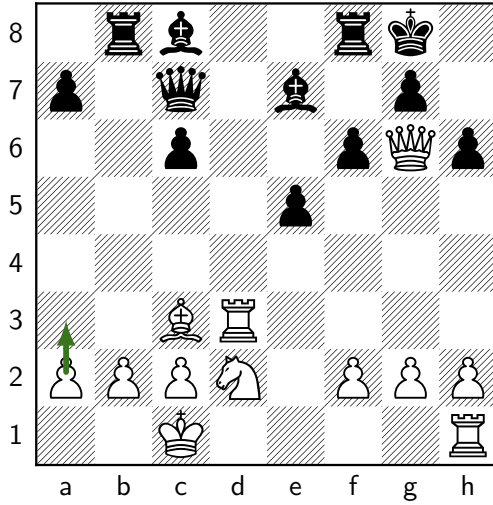
Figure 15: **Move 21, White: Bg5+.** (a) Our position; Melanie plays Bg5+ (c1g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#257` (ply 56) [structural distance $d=77$, QAP obj=90]: the master played h3; under μ our *Nb1* plays the role of that game’s *Ph2* (a **cross-type** correspondence); cross-type roles: our *Ra1* as their *Pg2*, our *Qd5* as their *Be3*; transferred move b1a3. (c) Retrieved from `ccr1_top_p0002#3741` (ply 37) [structural distance $d=61$, QAP obj=65]: the master played Bd5; under μ our *Re1* plays the role of that game’s *Bc3* (a **cross-type** correspondence); cross-type roles: our *pg7* as their *Ra8*, our *Bc1* as their *Pf7*; transferred move e1e3. (d) Retrieved from `ccr1_top_p0000#944` (ply 40) [structural distance $d=75$, QAP obj=61]: the master played c5; under μ our *Pb2* plays the role of that game’s *Pc4*; cross-type roles: our *Pa2* as their *Rb1*, our *Nb1* as their *Pa2*; transferred move b2b3. Synthesis: no retrieved analog transferred to Bg5+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



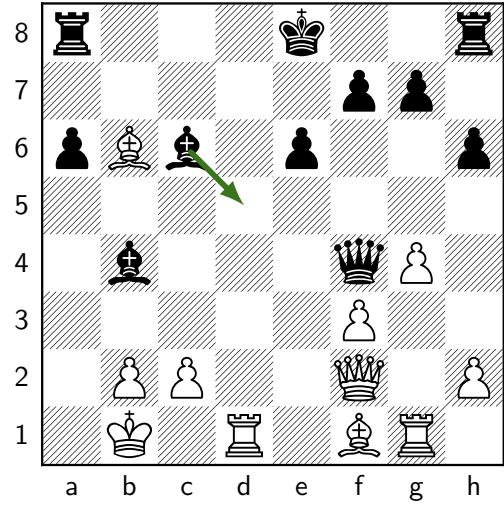
(a)



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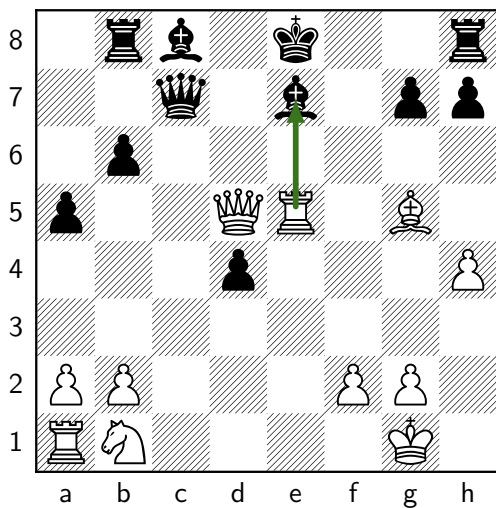


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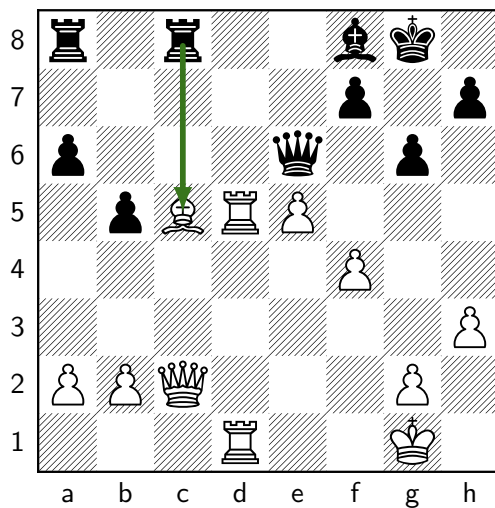


(d)

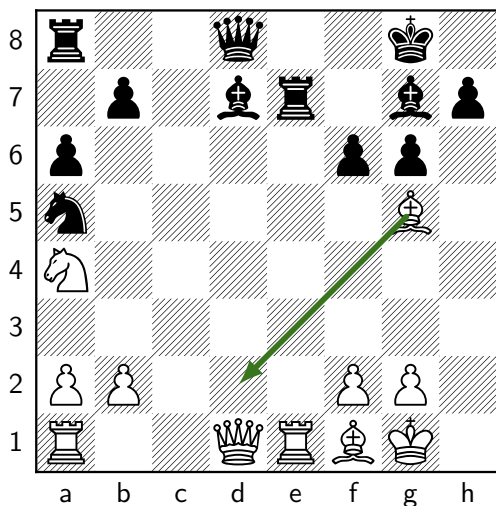
Figure 16: **Move 22, White: Rxe5+.** (a) Our position; Melanie plays Rxe5+ (e1e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0000#257` (ply 56) [structural distance $d=77$, QAP obj=113]: the master played h3; under μ our *Nb1* plays the role of that game’s *Ph2* (a **cross-type** correspondence); cross-type roles: our *Bg5* as their *Ra2*, our *Ra1* as their *Pg2*; transferred move b1d2. (c) Retrieved from `ccr1_top.p0000#761` (ply 38) [structural distance $d=63$, QAP obj=92]: the master played a3; under μ our *Bg5* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *rh8* as their *Pc2*, our *Qd5* as their *Bc3*; transferred move g5d2. (d) Retrieved from `ccr1_top.p0002#3741` (ply 37) [structural distance $d=60$, QAP obj=87]: the master played Bd5; under μ our *Re1* plays the role of that game’s *Bc3* (a **cross-type** correspondence); cross-type roles: our *Pa2* as their *Rh8*, our *Nb1* as their *Pf7*; transferred move e1e3. Synthesis: no retrieved analog transferred to Rxe5+ – the **search selector** chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



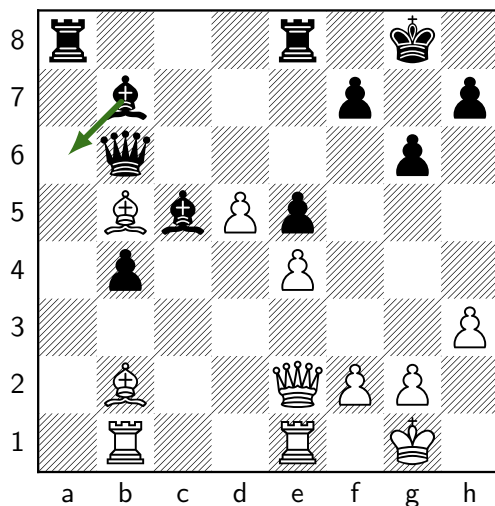
(a)



(b)

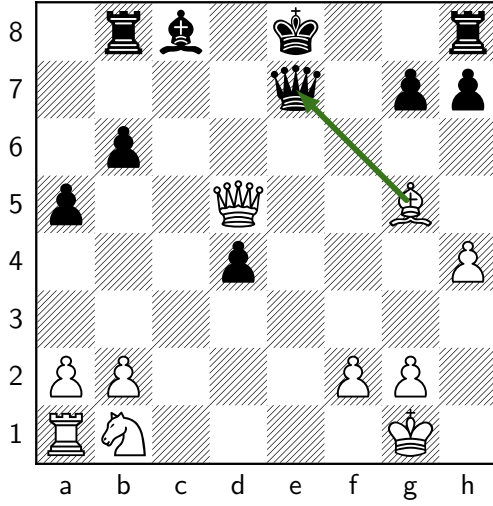


(c)

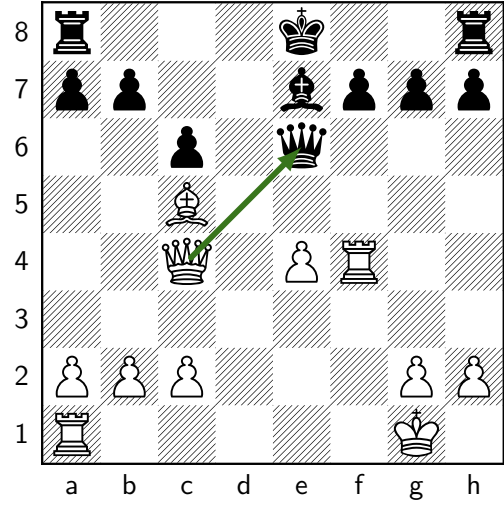


(d)

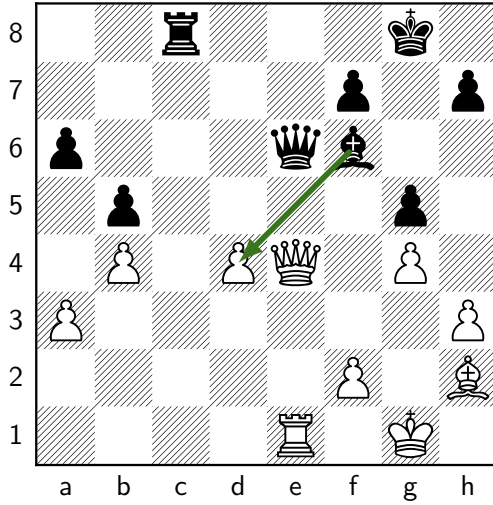
Figure 17: **Move 23, White: Rxe7+.** (a) Our position; Melanie plays Rxe7+ (e5e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#4000` (ply 47) [structural distance $d=84$, QAP obj=48]: the master played Rxc5; under μ our Re5 plays the role of that game’s Rc1; cross-type roles: our Qd5 as their Ra8, our rb8 as their Pa6; transferred move e5e3. (c) Retrieved from `ccr1_top_p0002#3244` (ply 40) [structural distance $d=78$, QAP obj=42]: the master played Bd2; under μ our Re5 plays the role of that game’s Bg5 (a **cross-type** correspondence); cross-type roles: our pb6 as their Ra1, our Re5 as their Qd1; transferred move e5e4. (d) Retrieved from `ccr1_top_p0001#3440` (ply 51) [structural distance $d=81$, QAP obj=37]: the master played Ba6; under μ our Bg5 plays the role of that game’s Bb2; cross-type roles: our pa5 as their Ra8, our be7 as their Re8; transferred move g5e3. Synthesis: no retrieved analog transferred to Rxe7+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



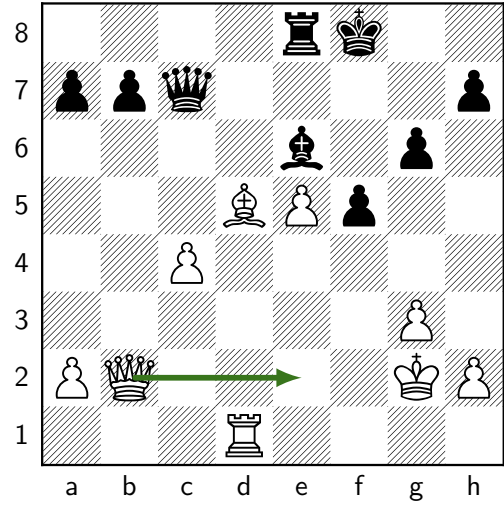
(a)



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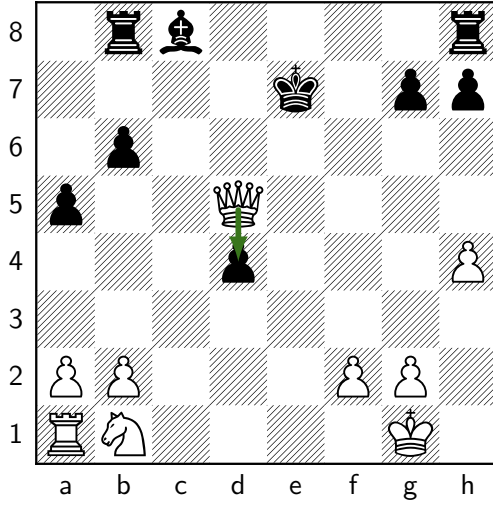


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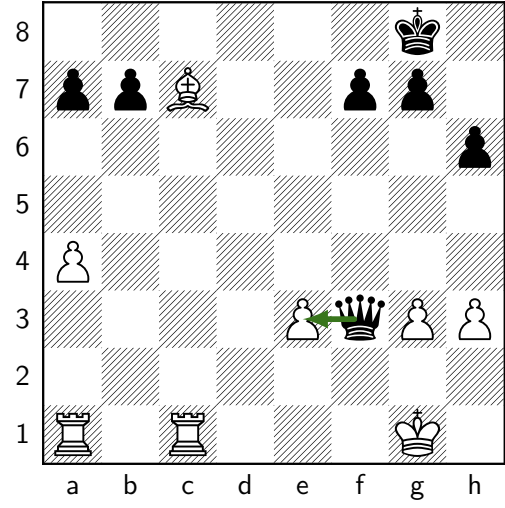


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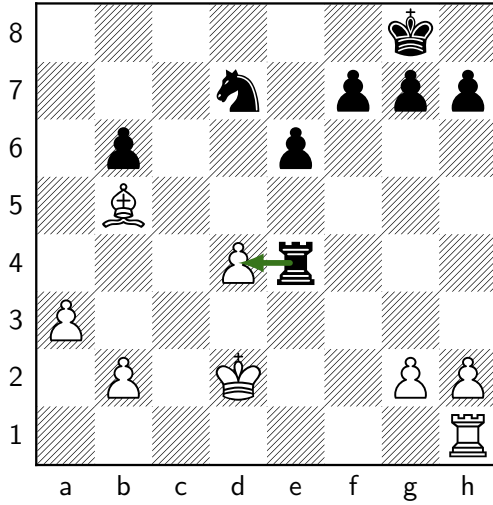
Figure 18: **Move 24, White: Bxe7**. (a) Our position; Melanie plays Bxe7 (g5e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1460` (ply 34) [structural distance $d=62$, QAP obj=34]: the master played Qxe6; under μ our Bg5 plays the role of that game's Qc4 (a **cross-type** correspondence); cross-type roles: our Nb1 as their Pa2, our bc8 as their Pe4; transferred move g5f6. (c) Retrieved from `ccr1_top_p0002#3341` (ply 57) [structural distance $d=70$, QAP obj=30]: the master played Bxd4; under μ our Pa2 plays the role of that game's Bf3 (a **cross-type** correspondence); cross-type roles: our Nb1 as their Ph7, our Bg5 as their Qe6; transferred move a2a3. (d) Retrieved from `ccr1_top_p0003#2662` (ply 58) [structural distance $d=68$, QAP obj=22]: the master played Qe2; under μ our Pf2 plays the role of that game's Qb2 (a **cross-type** correspondence); cross-type roles: our Nb1 as their Rd1, our Ra1 as their Pa2; transferred move f2f4. Synthesis: no retrieved analog transferred to Bxe7 – the **search selector** chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



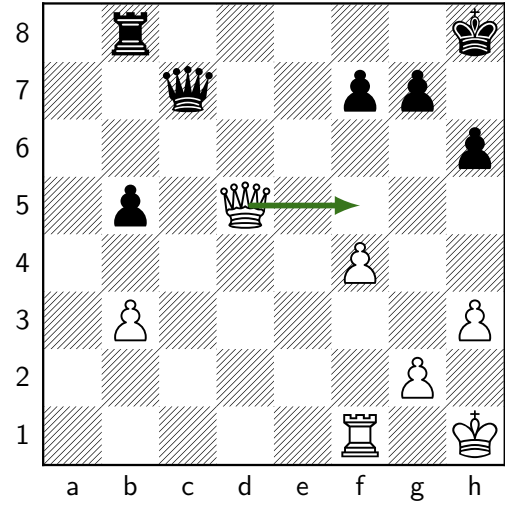
(a)



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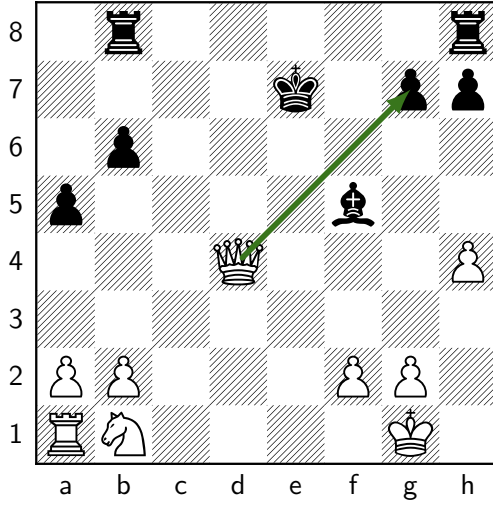


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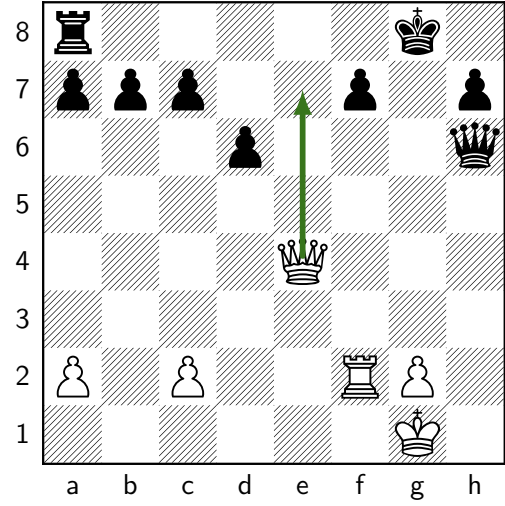


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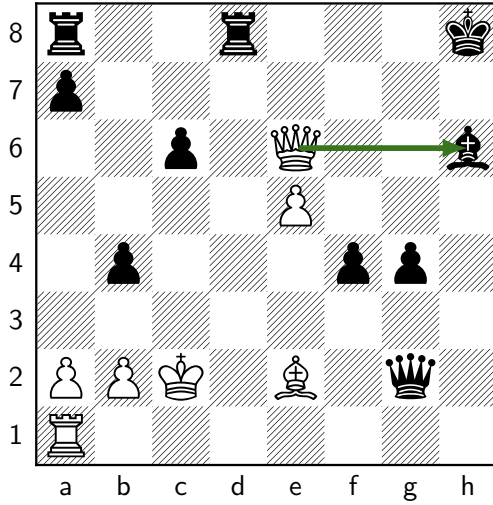
Figure 19: **Move 25, White: Qxd4.** (a) Our position; Melanie plays Qxd4 (d5d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2793` (ply 55) [structural distance $d=46$, QAP obj=38]: the master played Qxe3+; under μ our Qd5 plays the role of that game’s Qf6; cross-type roles: our Ra1 as their Pa7, our pb6 as their bc7; transferred move d5d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1910` (ply 47) [structural distance $d=72$, QAP obj=18]: the master played Rxd4+; under μ our Qd5 plays the role of that game’s Re5 (a **cross-type** correspondence); cross-type roles: our pd4 as their Nd7, our Nb1 as their Ph7; transferred move d5d4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1097` (ply 60) [structural distance $d=72$, QAP obj=22]: the master played Qf5; under μ our Qd5 plays the role of that game’s Qd5; cross-type roles: our Nb1 as their Rf1, our bc8 as their Ph3; transferred move d5c4. Synthesis: 3 retrieved analogs transferred to Qxd4 (summed QAP objective 79, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



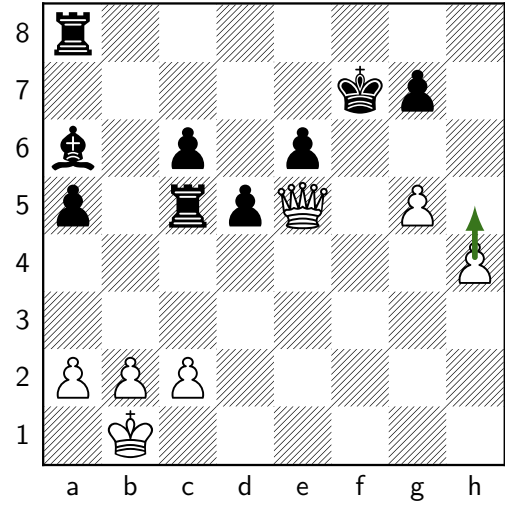
(a)



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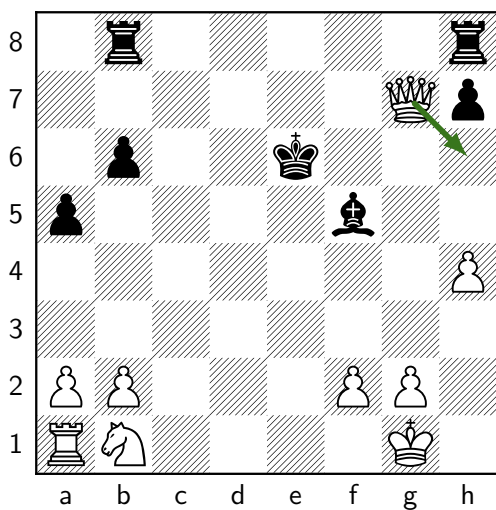


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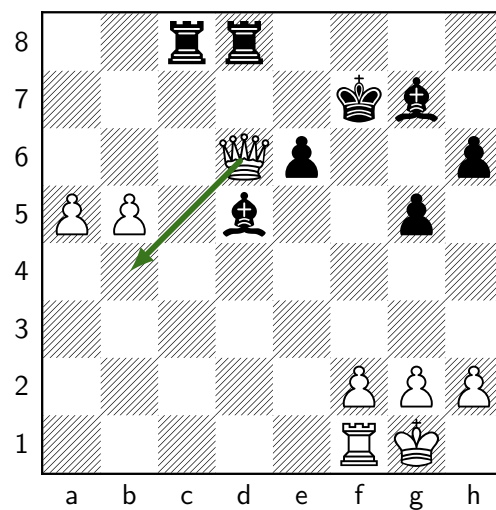


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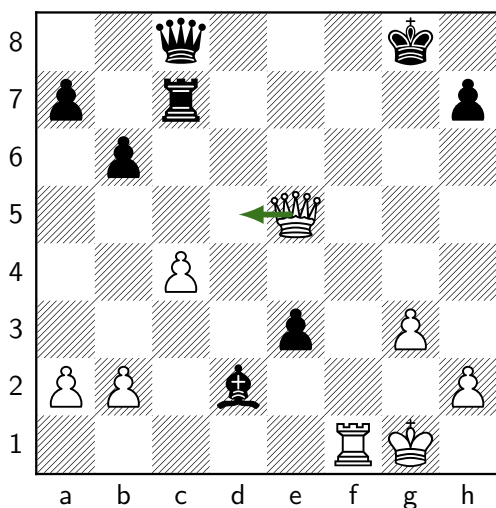
Figure 20: **Move 26, White: Qxg7+.** (a) Our position; Melanie plays Qxg7+ (d4g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#2287 (ply 52) [structural distance $d=76$, QAP obj=48]: the master played Qe7; under μ our Qd4 plays the role of that game’s Qe4; cross-type roles: our Ra1 as their Pa2, our bf5 as their Rf2; transferred move d4g7 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#1239 (ply 58) [structural distance $d=72$, QAP obj=24]: the master played Qxh6+; under μ our Qd4 plays the role of that game’s Qe6; cross-type roles: our Pg2 as their Ra1, our Ra1 as their Pb2; transferred move d4g7 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#401 (ply 64) [structural distance $d=70$, QAP obj=42]: the master played h5; under μ our Pg2 plays the role of that game’s Ph4; cross-type roles: our Nb1 as their Pc2, our bf5 as their rc5; transferred move g2g3. Synthesis: 2 retrieved analogs transferred to Qxg7+ (summed QAP objective 72, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



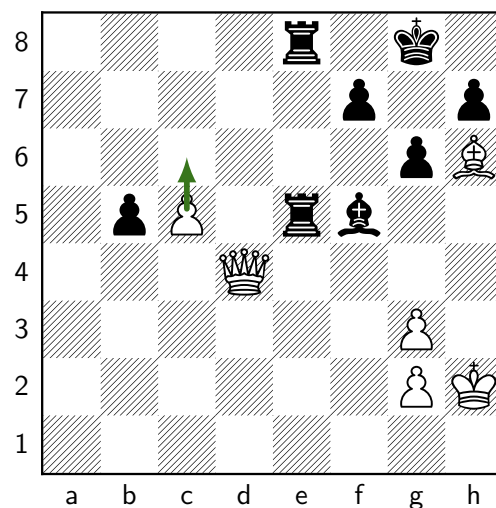
(a)



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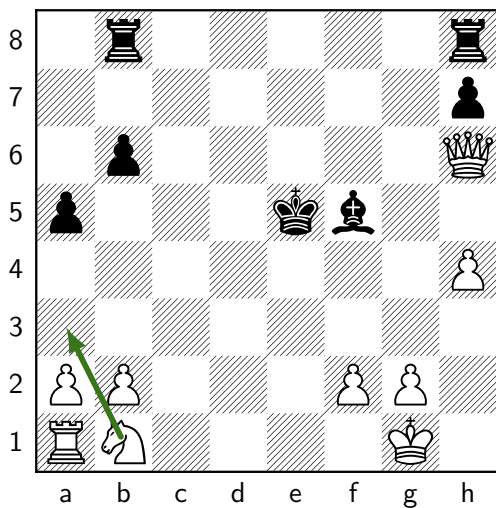


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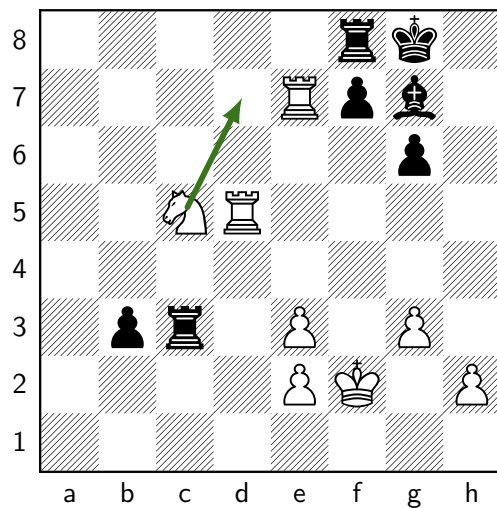


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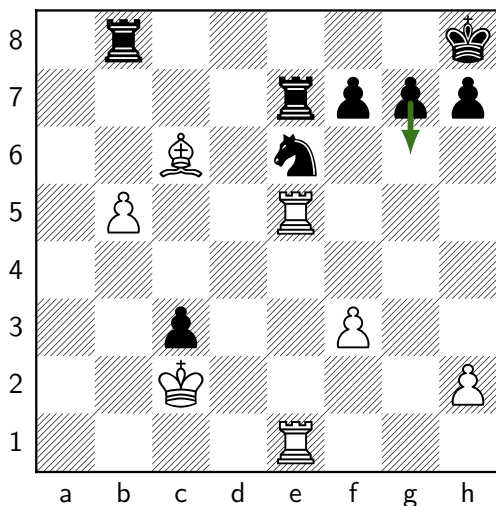
Figure 21: **Move 27, White: Qh6+.** (a) Our position; Melanie plays Qh6+ (g7h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#2088 (ply 62) [structural distance $d=45$, QAP obj=25]: the master played Qb4; under μ our Qg7 plays the role of that game’s Qd6; cross-type roles: our Pg2 as their Rf1, our Ra1 as their Pg2; transferred move g7h6 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#2765 (ply 62) [structural distance $d=54$, QAP obj=13]: the master played Qd5+; under μ our Qg7 plays the role of that game’s Qe5; cross-type roles: our pb6 as their Rf1, our Ra1 as their Ph2; transferred move g7h6 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#2195 (ply 66) [structural distance $d=57$, QAP obj=22]: the master played c6; under μ our Pb2 plays the role of that game’s Pc5; cross-type roles: our Nb1 as their Pg2, our rb8 as their bf5; transferred move b2b3. Synthesis: 2 retrieved analogs transferred to Qh6+ (summed QAP objective 38, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



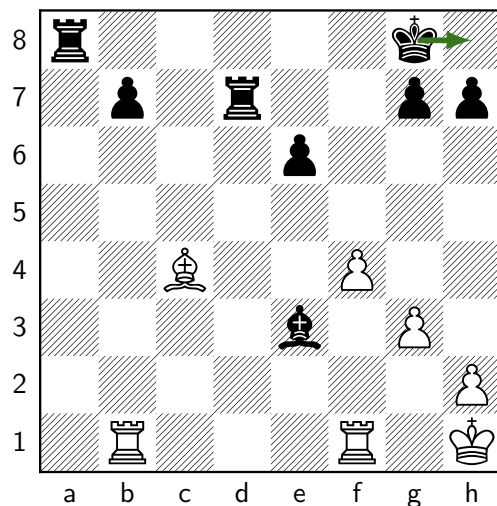
(a)



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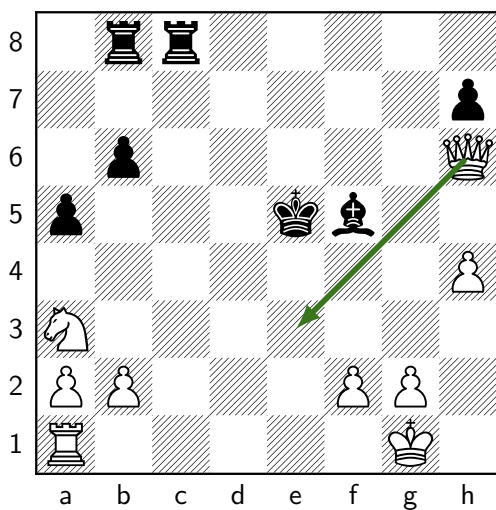


(c)

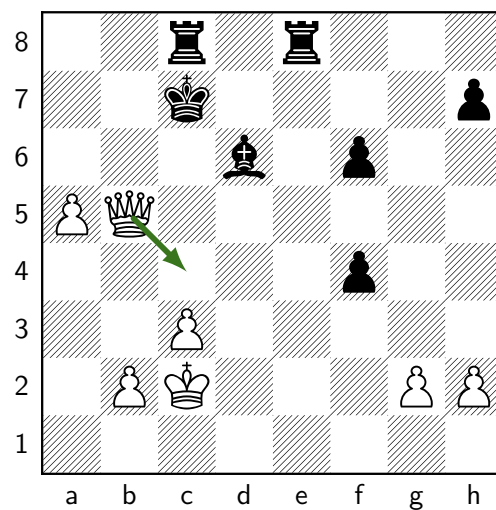


(d)

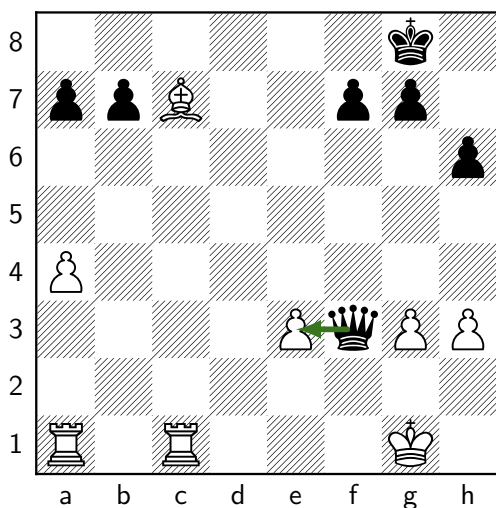
Figure 22: **Move 28, White: Na3**. (a) Our position; Melanie plays Na3 (b1a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1981` (ply 68) [structural distance $d=58$, QAP obj=45]: the master played Nd7; under μ our Nb1 plays the role of that game’s Nc5; cross-type roles: our bf5 as their rc3, our Ra1 as their Pg3; transferred move b1a3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#138` (ply 69) [structural distance $d=53$, QAP obj=17]: the master played g6; under μ our Nb1 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our bf5 as their Pf7, our Pg2 as their bc6; transferred move b1a3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#677` (ply 55) [structural distance $d=51$, QAP obj=34]: the master played Kh8; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Qh6 as their bc4, our Nb1 as their pf4; transferred move g1f1. Synthesis: 2 retrieved analogs transferred to Na3 (summed QAP objective 63, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



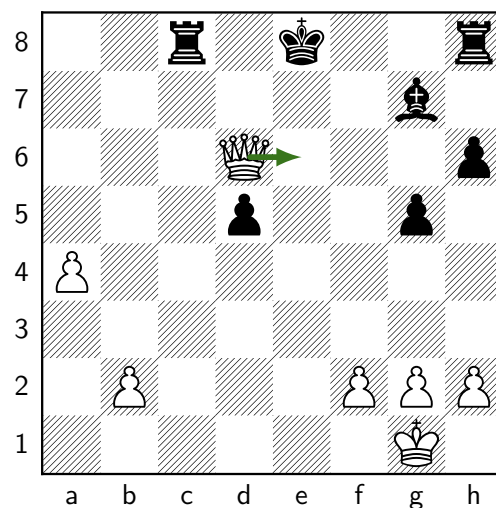
(a)



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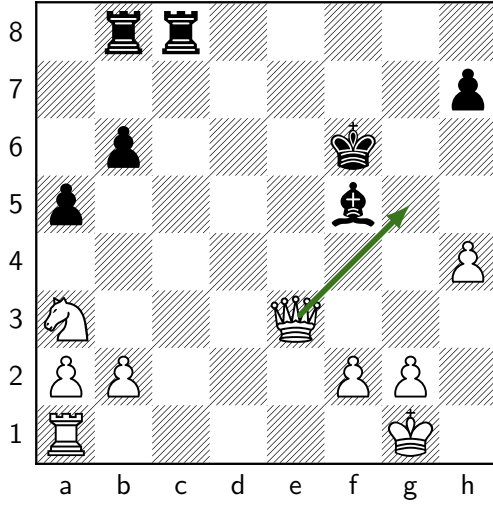


(c)

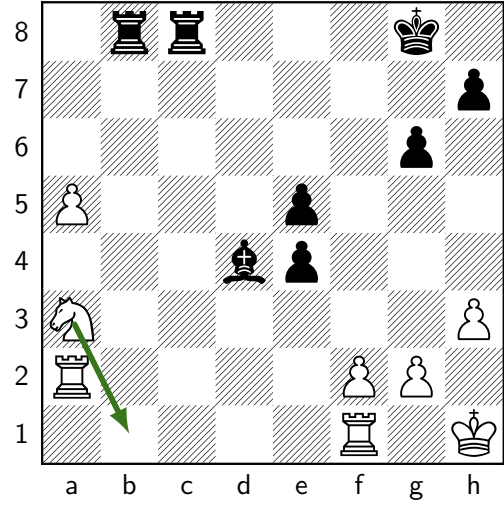


(d)

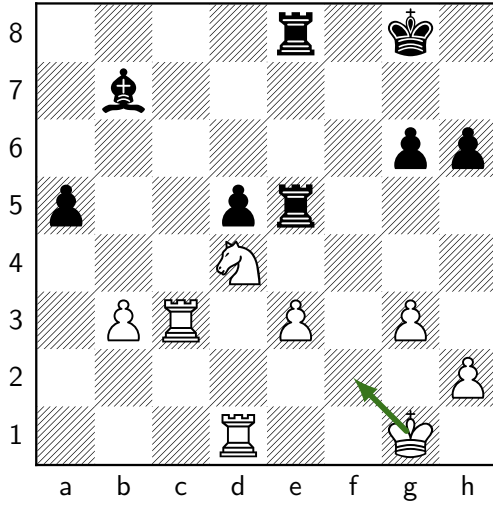
Figure 23: **Move 29, White: Qe3+.** (a) Our position; Melanie plays Qe3+ (h6e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1081` (ply 54) [structural distance $d=53$, QAP obj=14]: the master played Qc4+; under μ our Qh6 plays the role of that game’s Qb5; cross-type roles: our Na3 as their Ph2, our rc8 as their pf4; transferred move h6e3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2793` (ply 55) [structural distance $d=45$, QAP obj=37]: the master played Qxe3+; under μ our Qh6 plays the role of that game’s Qf6; cross-type roles: our Na3 as their Pa7, our Ra1 as their Pg7; transferred move h6g7. (d) Retrieved from `ccr1_top_p0000#520` (ply 60) [structural distance $d=52$, QAP obj=23]: the master played Qe6+; under μ our Qh6 plays the role of that game’s Qd6; cross-type roles: our Na3 as their Pb2, our Ra1 as their Pf2; transferred move h6c6. Synthesis: 1 retrieved analog transferred to Qe3+ (summed QAP objective 14, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



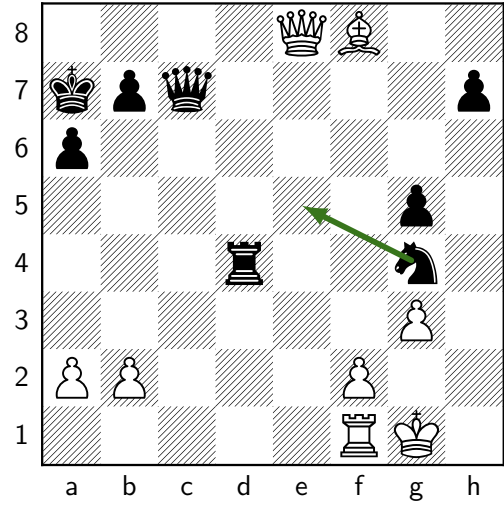
(a)



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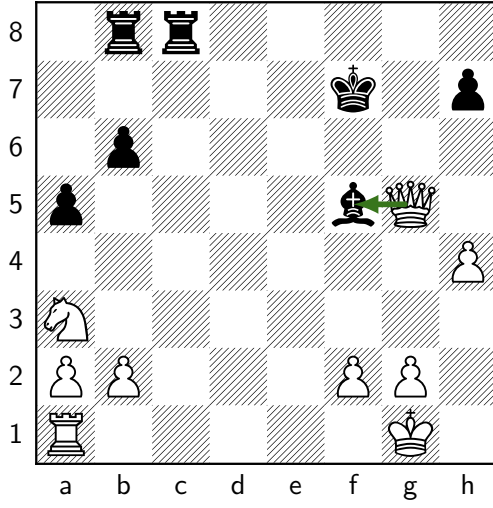


(c)

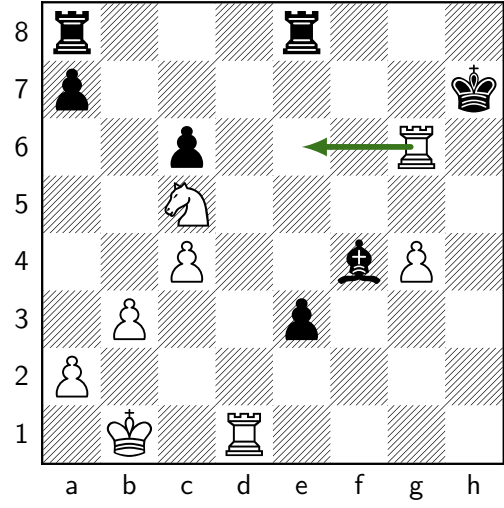


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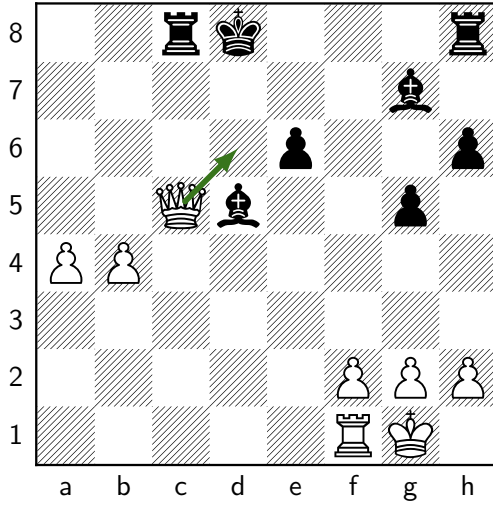
Figure 24: **Move 30, White: Qg5+.** (a) Our position; Melanie plays Qg5+ (e3g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#776` (ply 72) [structural distance $d=56$, QAP obj=17]: the master played Nb1; under μ our Pb2 plays the role of that game’s Na3 (a **cross-type** correspondence); cross-type roles: our bf5 as their Ra2, our Qe3 as their bd4; transferred move b2b3. (c) Retrieved from `ccr1_top_p0002#1243` (ply 58) [structural distance $d=60$, QAP obj=16]: the master played Kf2; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Na3 as their Ph2, our rc8 as their Pg3; transferred move g1h1. (d) Retrieved from `ccr1_top_p0000#3093` (ply 57) [structural distance $d=56$, QAP obj=16]: the master played Ne5; under μ our Qe3 plays the role of that game’s Ng5 (a **cross-type** correspondence); cross-type roles: our Pf2 as their qe8, our Na3 as their bf8; transferred move e3c3. Synthesis: no retrieved analog transferred to Qg5+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



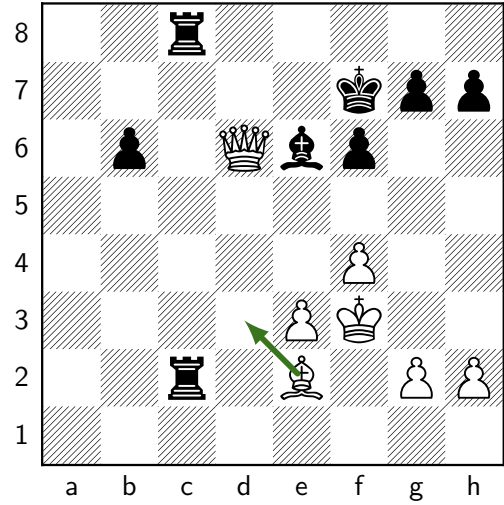
(a)



(b)

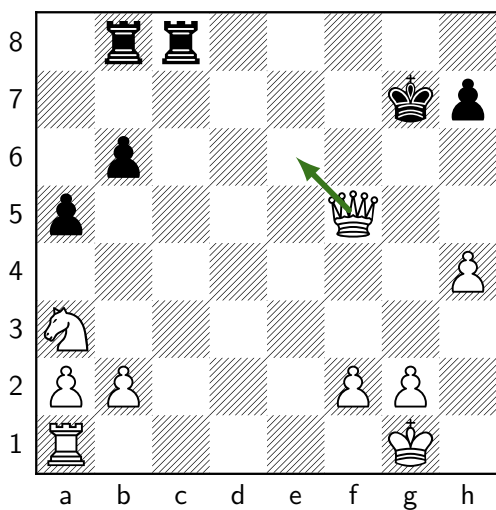


(c)

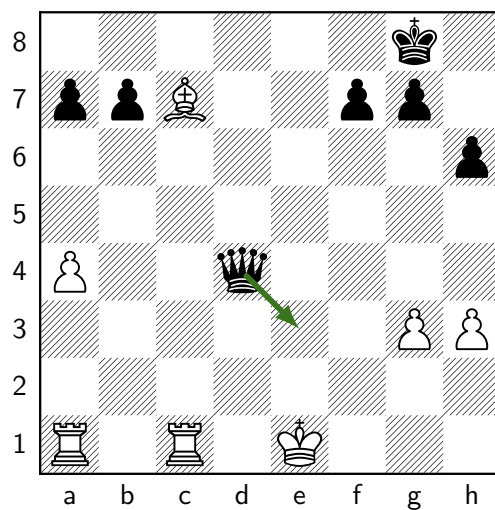


(d)

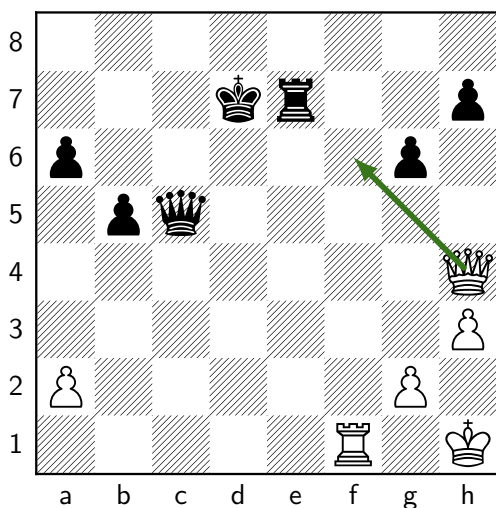
Figure 25: **Move 31, White: Qxf5+.** (a) Our position; Melanie plays Qxf5+ (g5f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1594` (ply 72) [structural distance $d=53$, QAP obj=2]: the master played R_e6 ; under μ our Q_g5 plays the role of that game’s R_g6 (a **cross-type** correspondence); cross-type roles: our $Na3$ as their P_b3 , our $ph7$ as their $bf4$; transferred move $g5f5$ (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2088` (ply 56) [structural distance $d=55$, QAP obj=27]: the master played Q_d6+ ; under μ our Q_g5 plays the role of that game’s Q_c5 ; cross-type roles: our P_b2 as their R_f1 , our $Na3$ as their P_g2 ; transferred move $g5f4$. (d) Retrieved from `ccr1_top_p0002#1754` (ply 50) [structural distance $d=65$, QAP obj=21]: the master played B_d3 ; under μ our $Na3$ plays the role of that game’s $Be2$ (a **cross-type** correspondence); cross-type roles: our $Ra1$ as their $pg7$; transferred move $a3b5$. Synthesis: 1 retrieved analog transferred to Qxf5+ (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



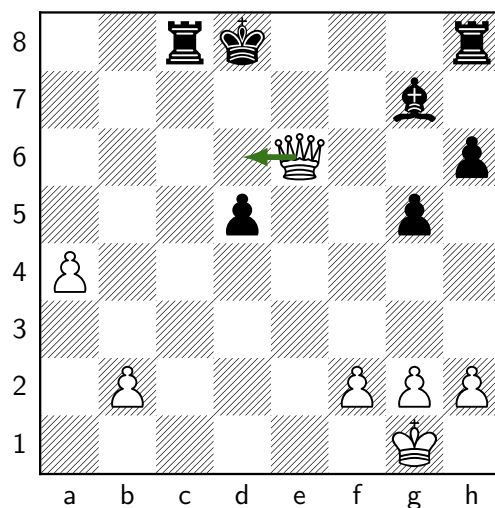
(a)



(b)

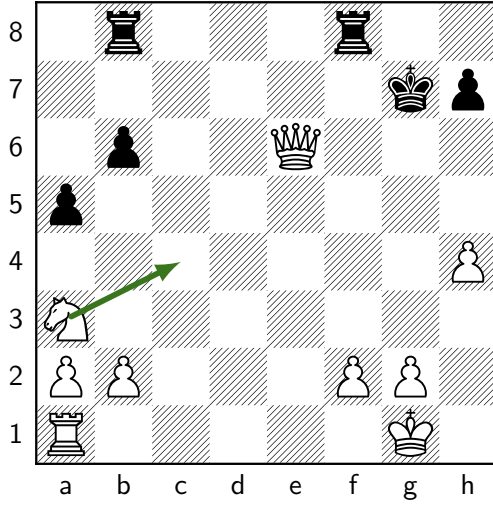


(c)

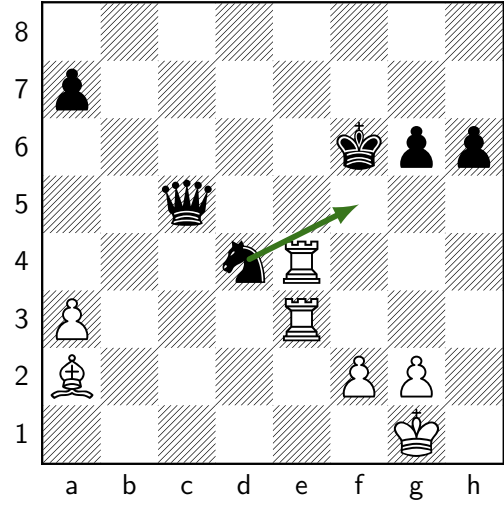


(d)

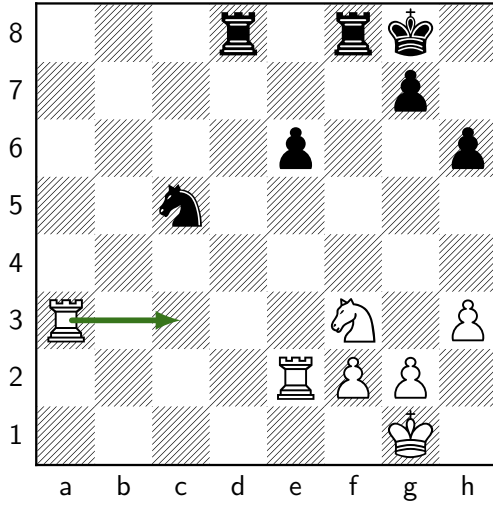
Figure 26: **Move 32, White: Qe6.** (a) Our position; Melanie plays Qe6 (f5e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2793` (ply 75) [structural distance $d=48$, QAP obj=27]: the master played Qe3+; under μ our Qf5 plays the role of that game's Qd5; cross-type roles: our Pg2 as their bc7, our Ra1 as their Pf7; transferred move f5e6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2543` (ply 70) [structural distance $d=53$, QAP obj=25]: the master played Qf6; under μ our Qf5 plays the role of that game's Qh4; cross-type roles: our Na3 as their Rf1, our Ra1 as their Pa2; transferred move f5e6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#520` (ply 62) [structural distance $d=50$, QAP obj=18]: the master played Qd6+; under μ our Qf5 plays the role of that game's Qe6; cross-type roles: our Na3 as their Pb2, our Ra1 as their Pf2; transferred move f5e6 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Qe6 (summed QAP objective 153, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



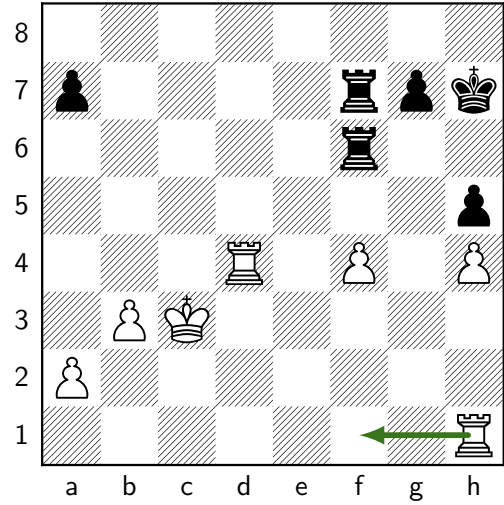
(a)



(b)

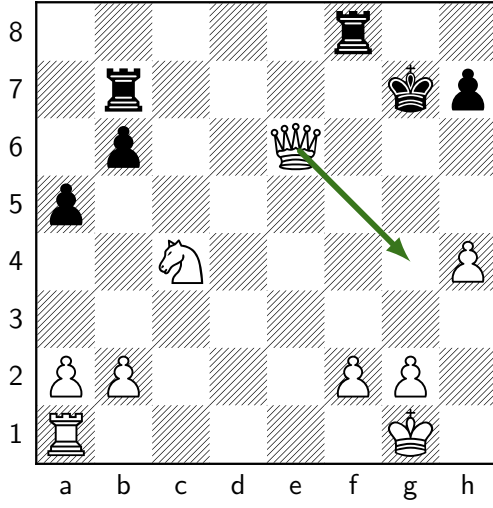


(c)

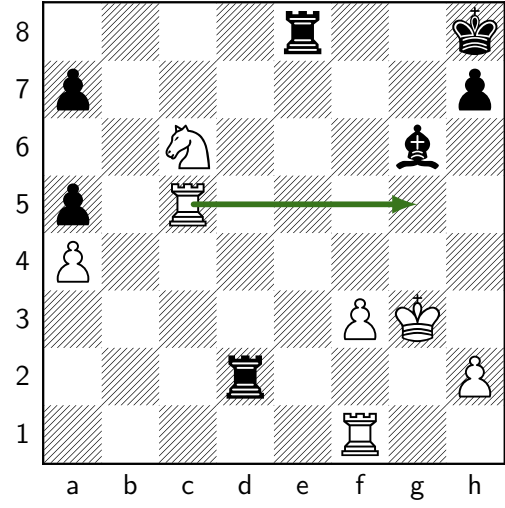


(d)

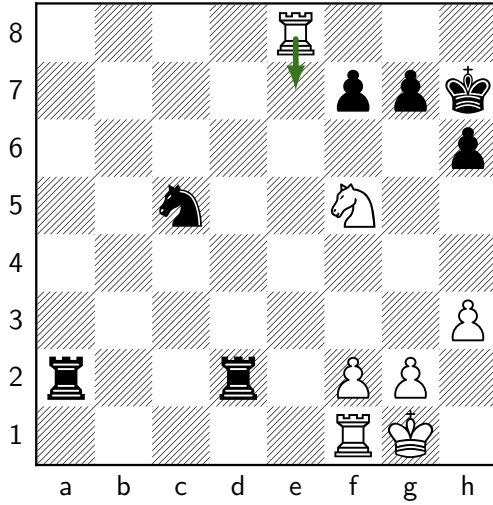
Figure 27: **Move 33, White: Nc4**. (a) Our position; Melanie plays Nc4 (a3c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2890` (ply 77) [structural distance $d=40$, QAP obj=2]: the master played Nf5; under μ our Na3 plays the role of that game’s Nd5; cross-type roles: our Ra1 as their Pa7, our ph7 as their re4; transferred move a3c4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1283` (ply 80) [structural distance $d=35$, QAP obj=16]: the master played Rc3; under μ our Pb2 plays the role of that game’s Ra3 (a **cross-type** correspondence); cross-type roles: our rb8 as their Pf2, our pb6 as their Nf3; transferred move b2b3. (d) Retrieved from `ccrl_top_p0001#602` (ply 70) [structural distance $d=38$, QAP obj=14]: the master played Rf1; under μ our Ra1 plays the role of that game’s Rh1; cross-type roles: our Na3 as their Pb3, our rf8 as their ph5; transferred move a1c1. Synthesis: 1 retrieved analog transferred to Nc4 (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



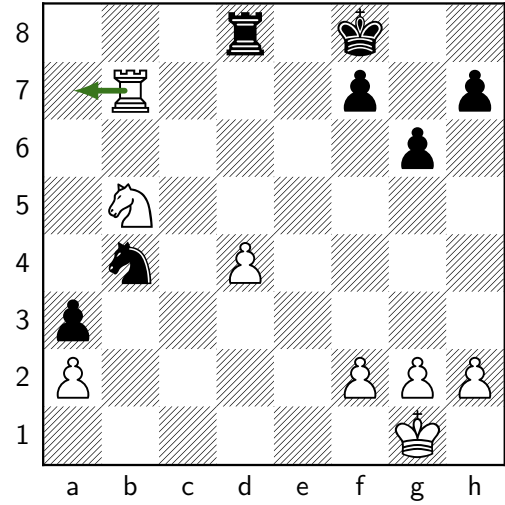
(a)



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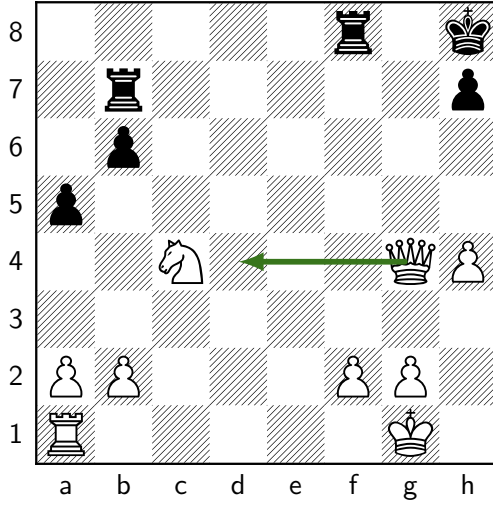


(c)

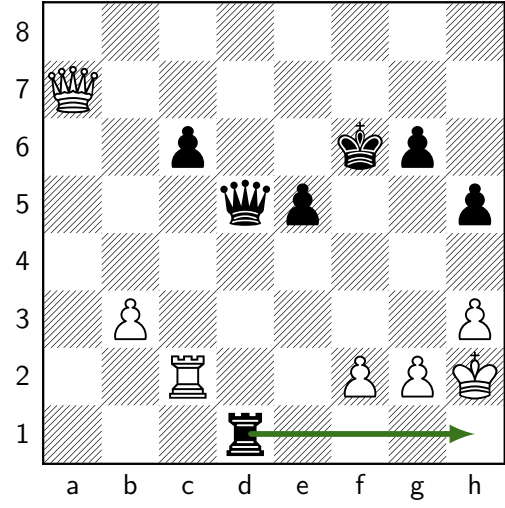


(d)

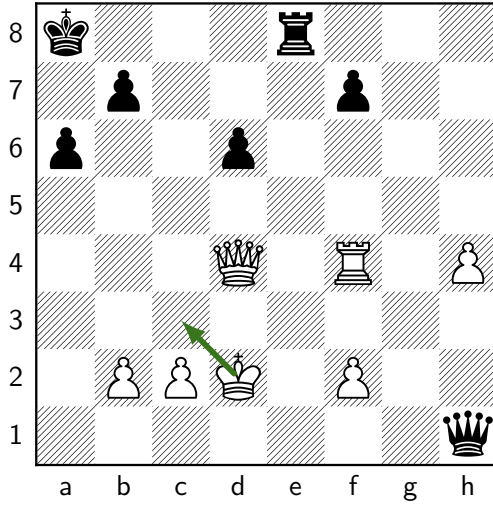
Figure 28: **Move 34, White: Qg4+.** (a) Our position; Melanie plays Qg4+ (e6g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#960` (ply 62) [structural distance $d=46$, QAP obj=72]: the master played Rg5; under μ our Qe6 plays the role of that game's Rc5 (a **cross-type** correspondence); cross-type roles: our Pg2 as their Rf1, our Ra1 as their Pa4; transferred move e6g4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2982` (ply 72) [structural distance $d=46$, QAP obj=65]: the master played Re7; under μ our Pa2 plays the role of that game's Re8 (a **cross-type** correspondence); cross-type roles: our Qe6 as their ra2, our Nc4 as their rd2; transferred move a2a3. (d) Retrieved from `ccr1_top_p0002#3106` (ply 54) [structural distance $d=46$, QAP obj=44]: the master played Ra7; under μ our Qe6 plays the role of that game's Rb7 (a **cross-type** correspondence); cross-type roles: our Ra1 as their pg6; transferred move e6e8. Synthesis: 1 retrieved analog transferred to Qg4+ (summed QAP objective 72, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



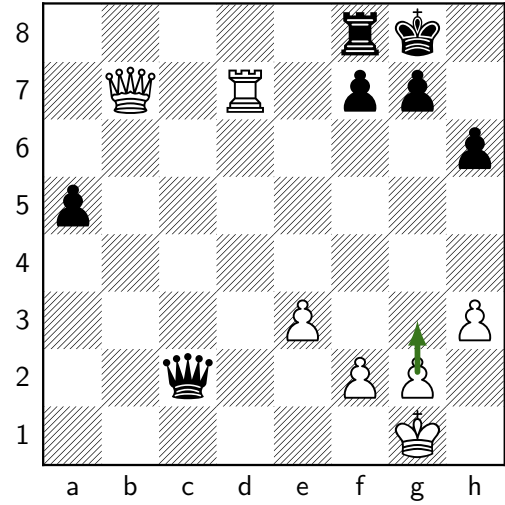
(a)



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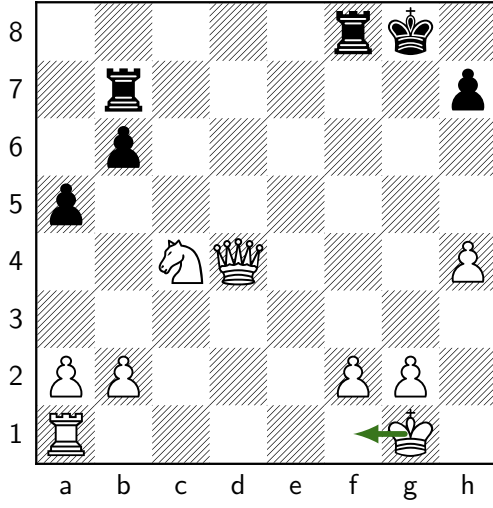


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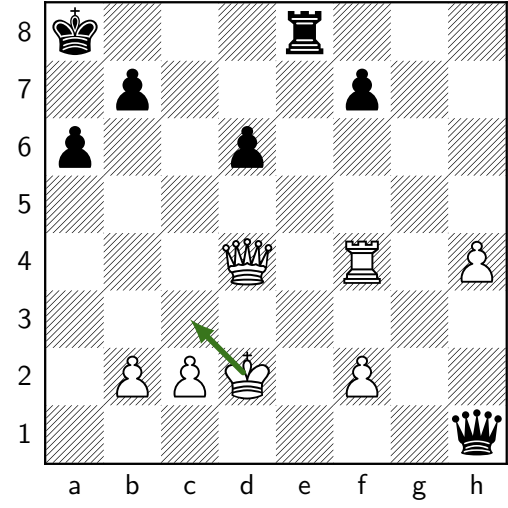


(d)

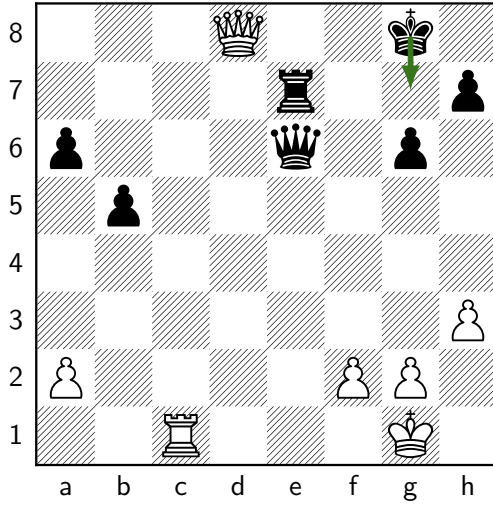
Figure 29: **Move 35, White: Qd4+.** (a) Our position; Melanie plays Qd4+ (g4d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3240` (ply 75) [structural distance $d=58$, QAP obj=40]: the master played Rh1+; under μ our Qg4 plays the role of that game’s Rd8 (a **cross-type** correspondence); cross-type roles: our Pg2 as their qa7, our Ra1 as their Pg6; transferred move g4g8. (c) Retrieved from `ccr1_top_p0001#16` (ply 70) [structural distance $d=48$, QAP obj=33]: the master played Kc3; under μ our Kg1 plays the role of that game’s Kd2; cross-type roles: our rf8 as their qh1, our Ra1 as their Pc2; transferred move g1f1. (d) Retrieved from `ccr1_top_p0002#662` (ply 54) [structural distance $d=58$, QAP obj=29]: the master played g3; under μ our Ph4 plays the role of that game’s Pg2; cross-type roles: our rf8 as their qc2, our Ra1 as their ph6; transferred move h4h5. Synthesis: no retrieved analog transferred to Qd4+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



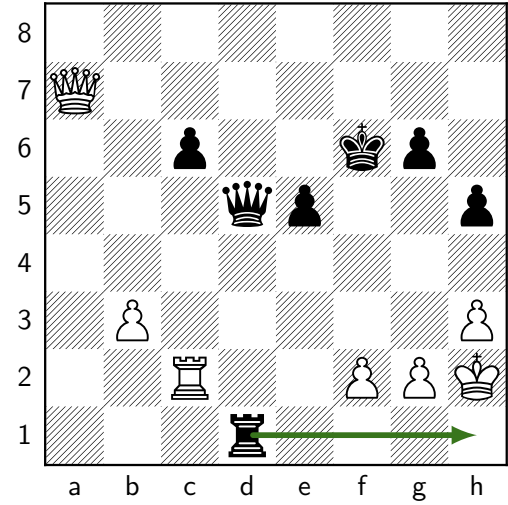
(a)



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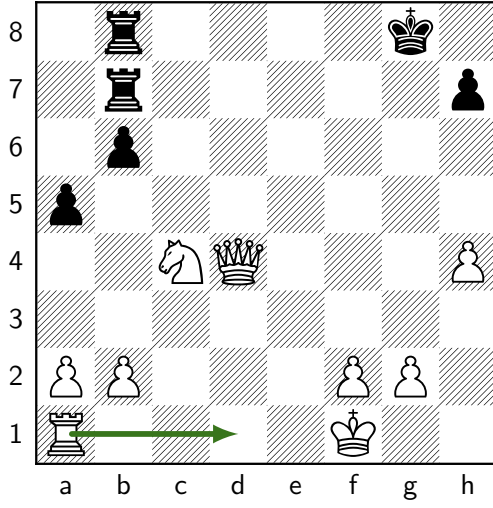


(c)

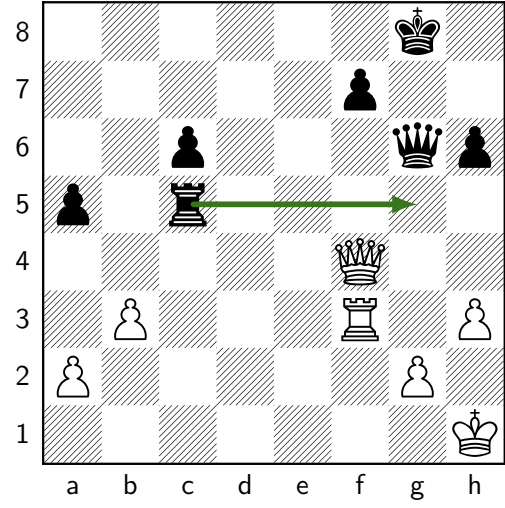


(d)

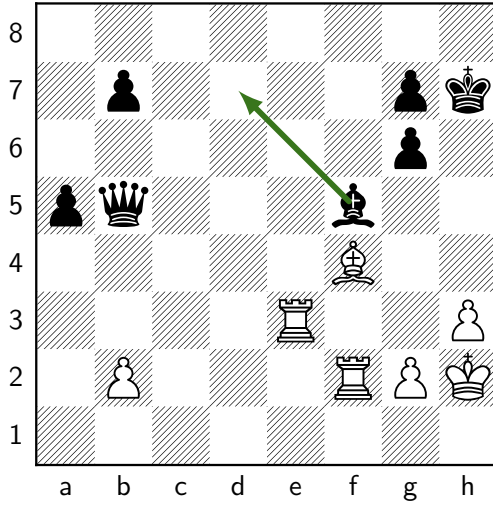
Figure 30: **Move 36, White: Kf1.** (a) Our position; Melanie plays Kf1 (g1f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#16` (ply 70) [structural distance $d=58$, QAP obj=40]: the master played Kc3; under μ our Kg1 plays the role of that game’s Kd2; cross-type roles: our rf8 as their qh1, our Ra1 as their Pb2; transferred move g1f1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#667` (ply 63) [structural distance $d=59$, QAP obj=4]: the master played Kg7; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our rb7 as their qd8, our Qd4 as their Re7; transferred move g1f1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3240` (ply 75) [structural distance $d=68$, QAP obj=42]: the master played Rh1+; under μ our Qd4 plays the role of that game’s Rd8 (a **cross-type** correspondence); cross-type roles: our rb7 as their qa7, our Nc4 as their Qd5; transferred move d4h8. Synthesis: 2 retrieved analogs transferred to Kf1 (summed QAP objective 44, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



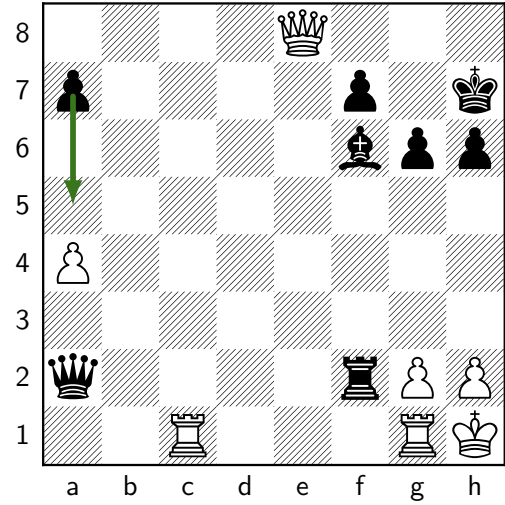
(a)



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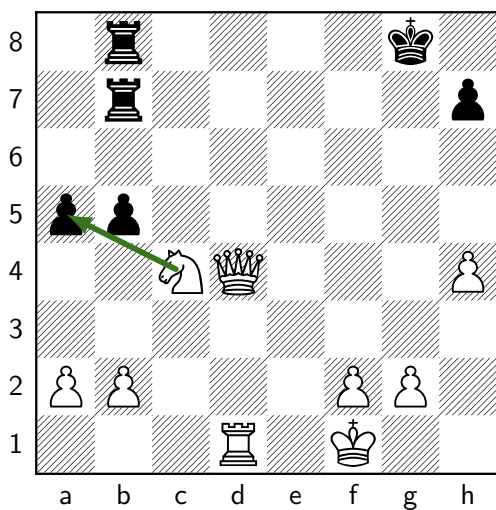


(c)

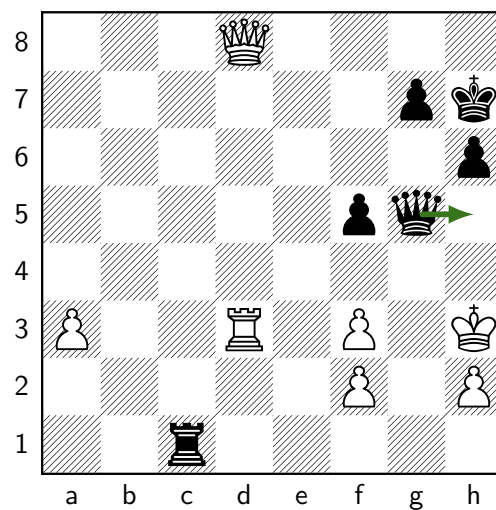


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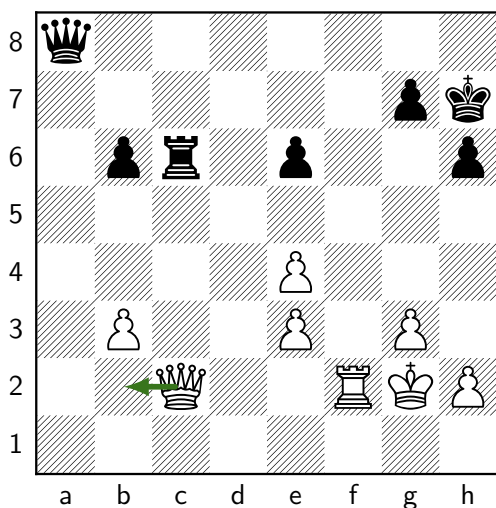
Figure 31: **Move 37, White: Rd1.** (a) Our position; Melanie plays Rd1 (a1d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1278` (ply 89) [structural distance $d=87$, QAP obj=45]: the master played Rg5; under μ our *Ra1* plays the role of that game’s *Rc4*; cross-type roles: our *rb8* as their *Pa5*, our *Nc4* as their *qf4*; transferred move **a1d1 (endorses this move)**. (c) Retrieved from `ccrl_top_p0001#2301` (ply 69) [structural distance $d=76$, QAP obj=36]: the master played Bd7; under μ our *Nc4* plays the role of that game’s *Bf4* (a **cross-type** correspondence); cross-type roles: our *rb8* as their *bf4*; transferred move **c4d2**. (d) Retrieved from `ccrl_top_p0000#3995` (ply 69) [structural distance $d=84$, QAP obj=23]: the master played a5; under μ our *Ra1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *Pf2* as their *qe8*, our *Ph4* as their *Bf6*; transferred move **a1c1**. Synthesis: 1 retrieved analog transferred to Rd1 (summed QAP objective 45, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



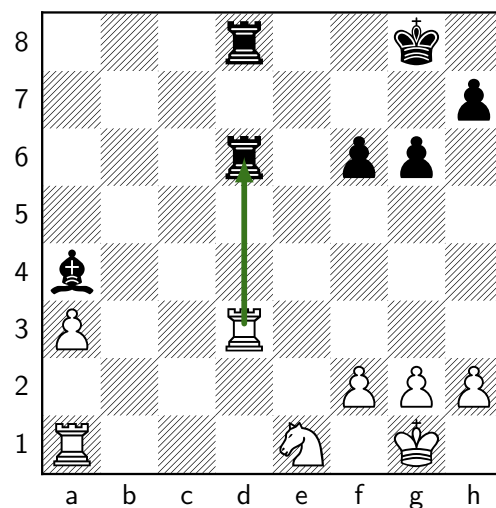
(a)



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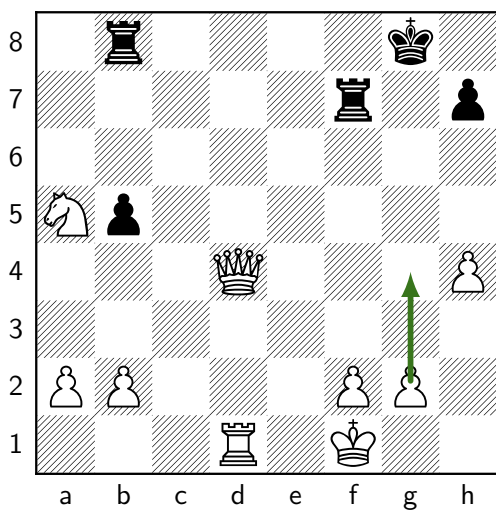


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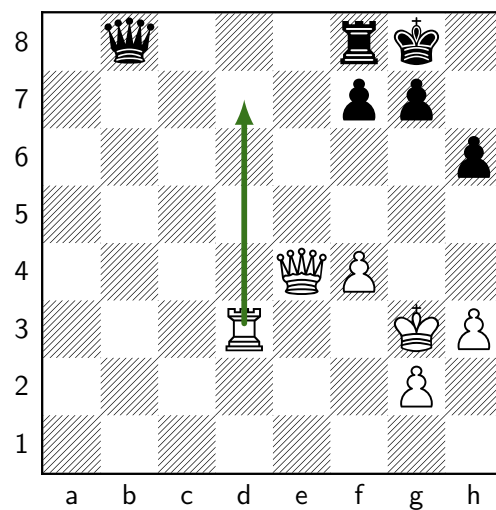


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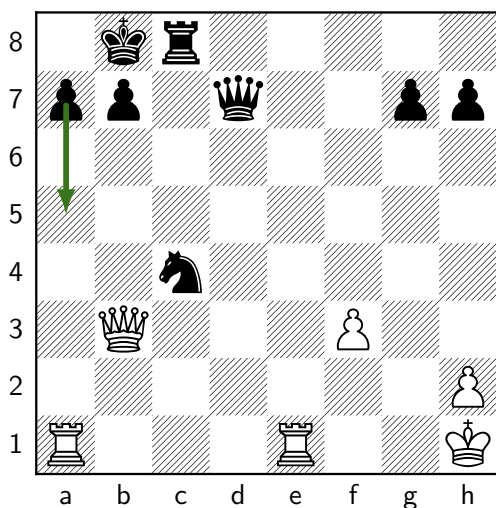
Figure 32: **Move 38, White: Nxa5.** (a) Our position; Melanie plays Nxa5 (c4a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#54` (ply 75) [structural distance $d=89$, QAP obj=19]: the master played Qh5+; under μ our Nc4 plays the role of that game’s Qg4 (a **cross-type** correspondence); cross-type roles: our pb5 as their qd8, our Qd4 as their Pf5; transferred move c4a5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3687` (ply 76) [structural distance $d=77$, QAP obj=16]: the master played Qb2; under μ our Nc4 plays the role of that game’s Qc2 (a **cross-type** correspondence); cross-type roles: our Qd4 as their Ph2, our rb8 as their pb6; transferred move c4a5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1458` (ply 60) [structural distance $d=85$, QAP obj=14]: the master played Rxd6; under μ our Nc4 plays the role of that game’s Rd3 (a **cross-type** correspondence); cross-type roles: our Pa2 as their Ra1, our Qd4 as their Ne1; transferred move c4a5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nxa5 (summed QAP objective 58, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



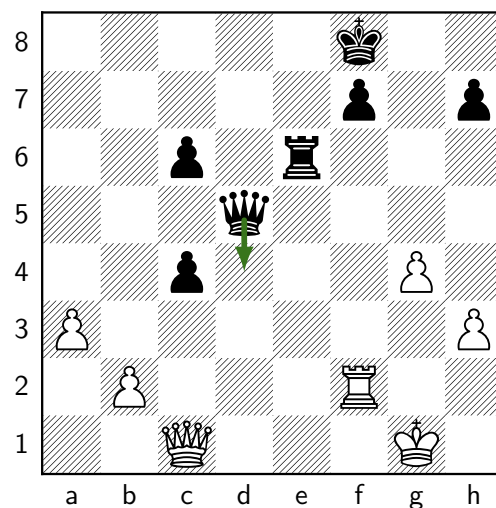
(a)



(b)

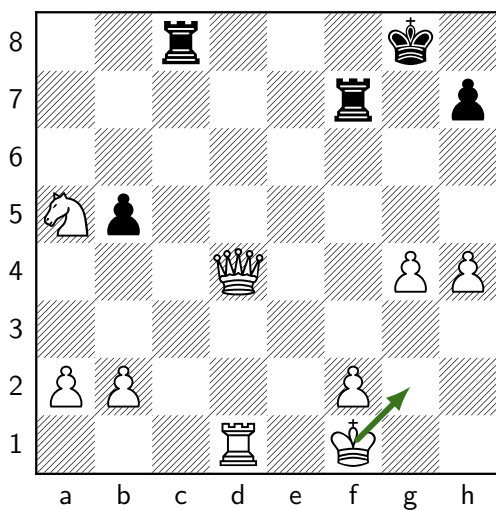


(c)

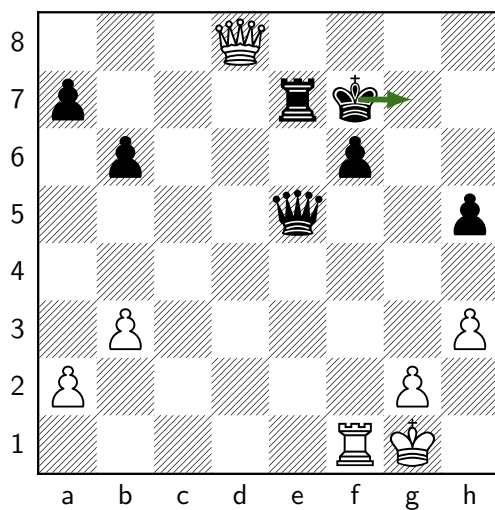


(d)

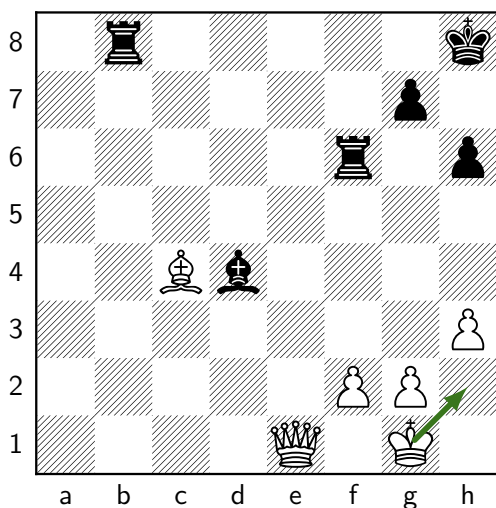
Figure 33: **Move 39, White: g4**. (a) Our position; Melanie plays g4 (g2g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2434` (ply 80) [structural distance $d=46$, QAP obj=38]: the master played Rd7; under μ our Pg2 plays the role of that game’s Rd3 (a **cross-type** correspondence); cross-type roles: our Rd1 as their Pg2, our rb8 as their pg7; transferred move g2g4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2264` (ply 77) [structural distance $d=43$, QAP obj=24]: the master played a5; under μ our Pg2 plays the role of that game’s Pa2; cross-type roles: our rf7 as their qb3, our Pa2 as their ra1; transferred move g2g4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1528` (ply 85) [structural distance $d=60$, QAP obj=44]: the master played Qd4; under μ our Qd4 plays the role of that game’s Qd4; cross-type roles: our Na5 as their Re6, our rb8 as their pg4; transferred move d4e5. Synthesis: 2 retrieved analogs transferred to g4 (summed QAP objective 63, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



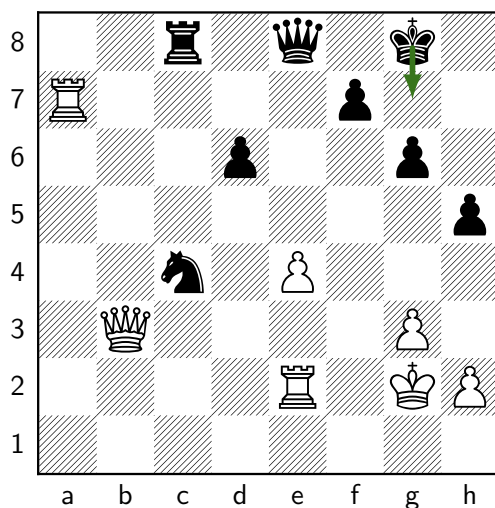
(a)



(b)

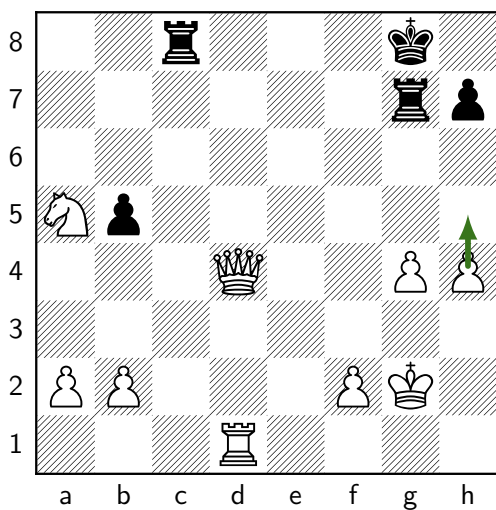


(c)

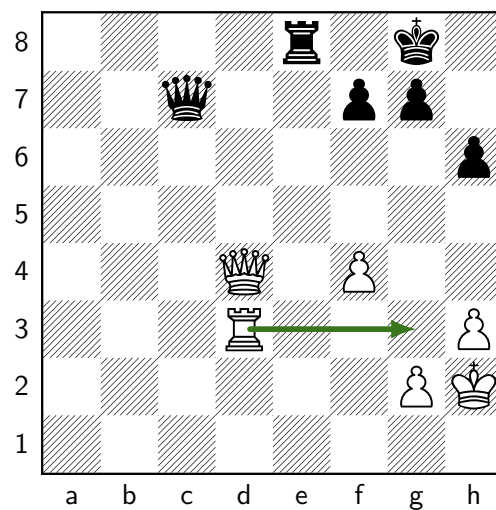


(d)

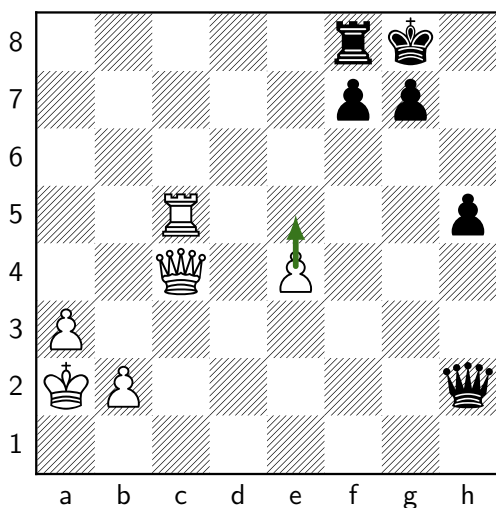
Figure 34: **Move 40, White: Kg2.** (a) Our position; Melanie plays Kg2 (f1g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1841` (ply 75) [structural distance $d=51$, QAP obj=28]: the master played Kg7; under μ our *Kf1* plays the role of that game’s *Kf2*; cross-type roles: our *pb5* as their *qd8*, our *Na5* as their *Pa7*; transferred move f1g2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3088` (ply 72) [structural distance $d=59$, QAP obj=26]: the master played Kh2; under μ our *Kf1* plays the role of that game’s *Kg1*; cross-type roles: our *Na5* as their *Bc4*, our *rf7* as their *bd4*; transferred move f1g2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2016` (ply 83) [structural distance $d=58$, QAP obj=19]: the master played Kg7; under μ our *Kf1* plays the role of that game’s *Kg1*; cross-type roles: our *pb5* as their *Rc8*, our *rf7* as their *Qe8*; transferred move f1g2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kg2 (summed QAP objective 74, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



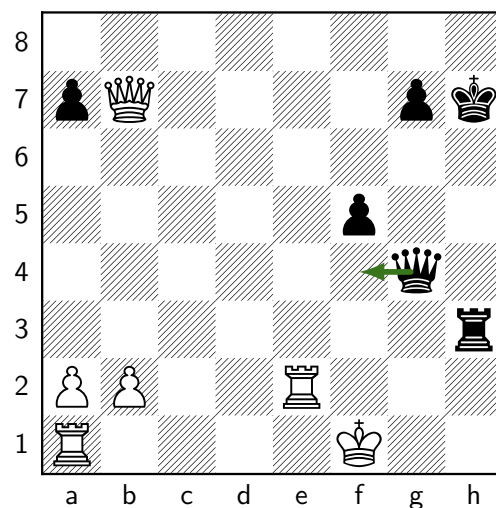
(a)



(b)

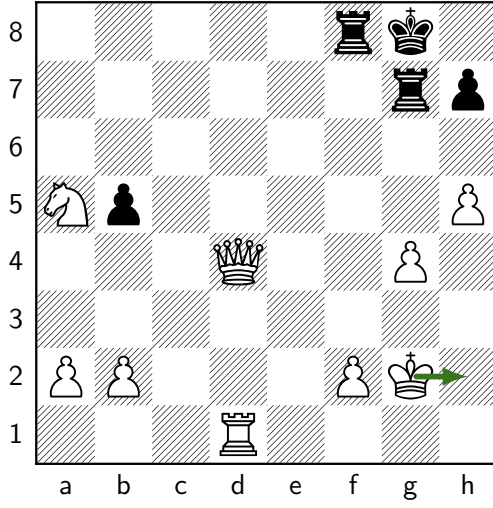


(c)

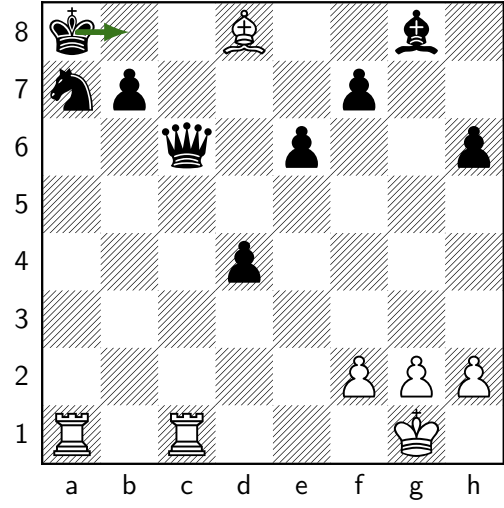


(d)

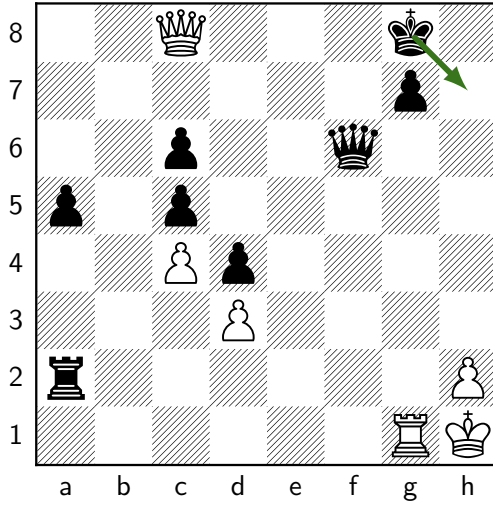
Figure 35: **Move 41, White: h5.** (a) Our position; Melanie plays h5 (h4h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2434` (ply 88) [structural distance $d=60$, QAP obj=25]: the master played Rg3; under μ our Ph4 plays the role of that game’s Rd3 (a **cross-type** correspondence); cross-type roles: our Rd1 as their Pg2, our Na5 as their ph6; transferred move h4h5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1093` (ply 82) [structural distance $d=60$, QAP obj=24]: the master played e5; under μ our Ph4 plays the role of that game’s Pe4; cross-type roles: our rg7 as their qh2, our Rd1 as their Pa3; transferred move h4h5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1197` (ply 85) [structural distance $d=55$, QAP obj=26]: the master played Qf4+; under μ our Qd4 plays the role of that game’s Qg5; cross-type roles: our rg7 as their qb7, our Rd1 as their Pf5; transferred move d4e5. Synthesis: 2 retrieved analogs transferred to h5 (summed QAP objective 49, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



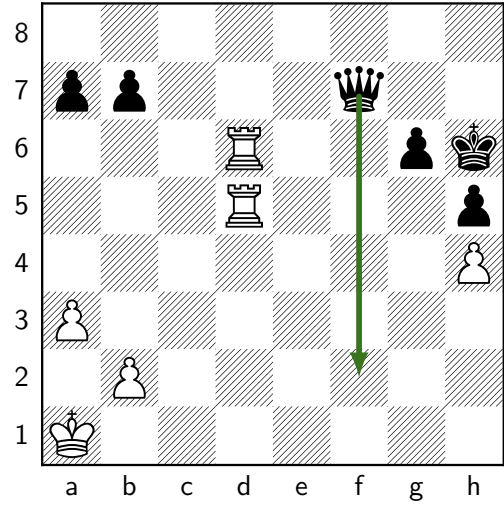
(a)



(b)



(c)



(d)

Figure 36: **Move 42, White: Kh2.** (a) Our position; Melanie plays Kh2 (g2h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#172` (ply 69) [structural distance $d=50$, QAP obj=27]: the master played Kb8; under μ our Kg2 plays the role of that game’s Ka1; cross-type roles: our ph7 as their bd8, our Rd1 as their Bg8; transferred move g2h2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2478` (ply 91) [structural distance $d=56$, QAP obj=17]: the master played Kh7; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our rf8 as their qc8, our Rd1 as their Pa5; transferred move g2h2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1924` (ply 89) [structural distance $d=52$, QAP obj=38]: the master played Qf2; under μ our Qd4 plays the role of that game’s Qf2; cross-type roles: our rf8 as their ph4, our Na5 as their pa3; transferred move d4b6. Synthesis: 2 retrieved analogs transferred to Kh2 (summed QAP objective 45, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

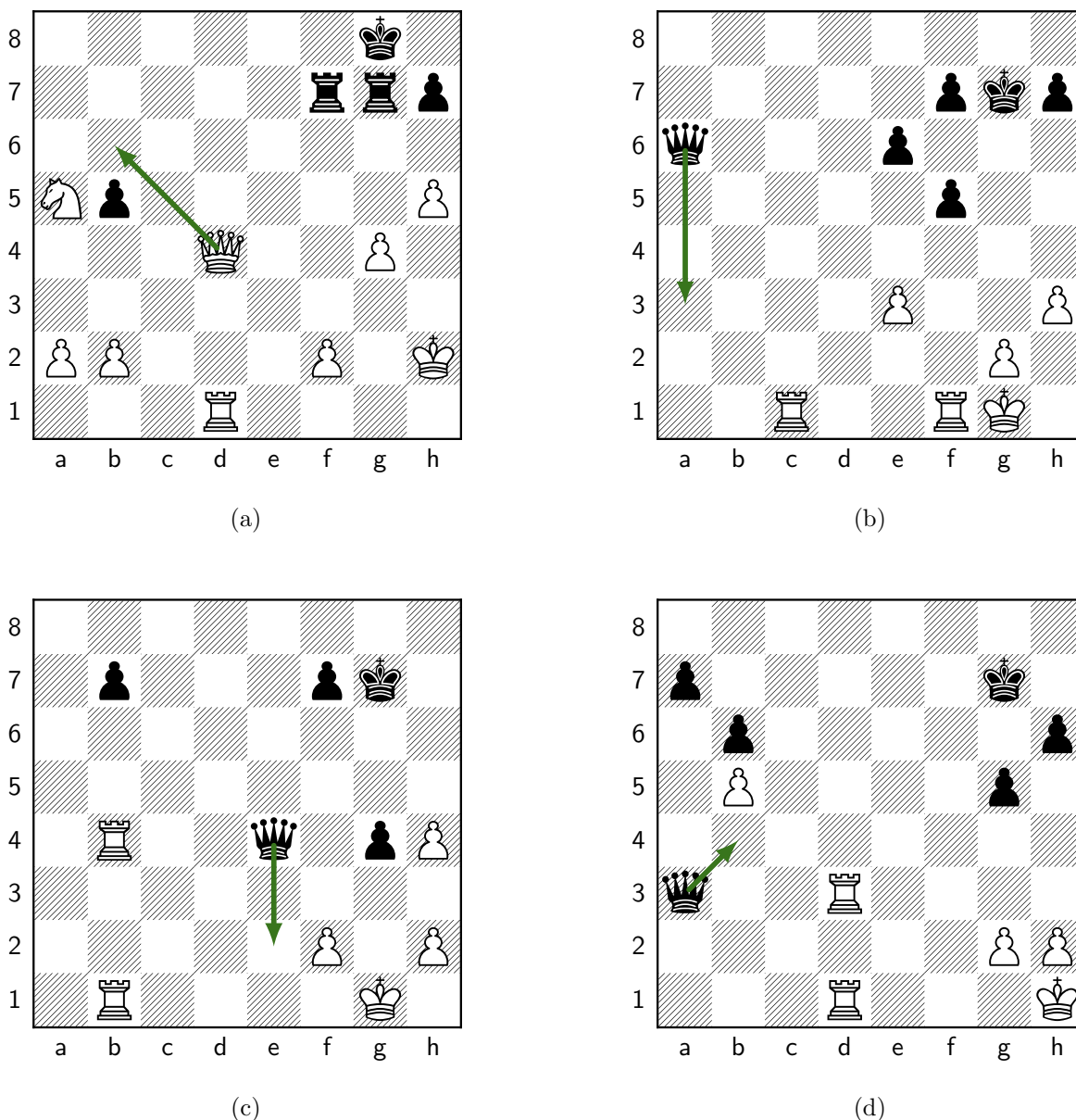
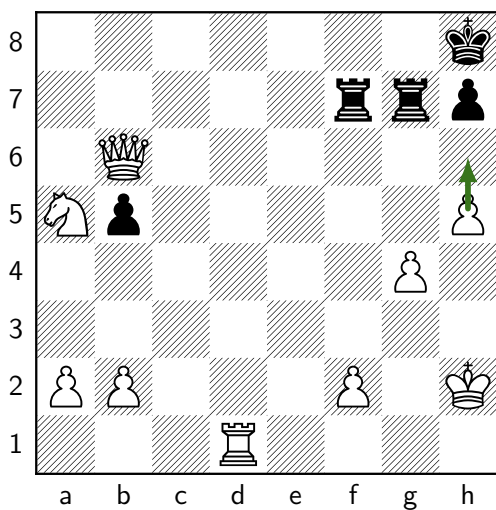
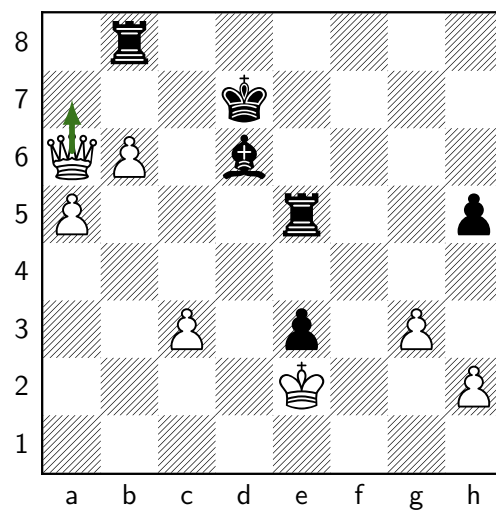


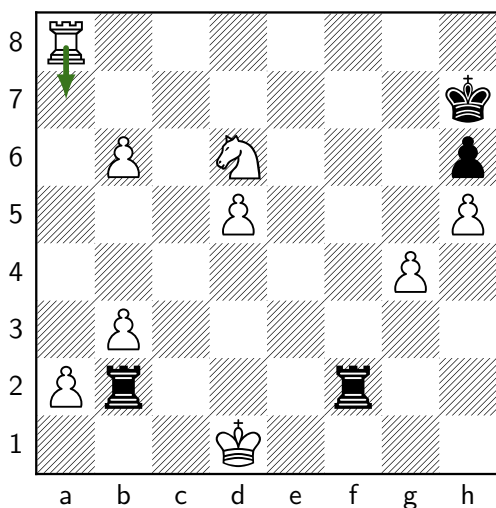
Figure 37: **Move 43, White: Qb6.** (a) Our position; Melanie plays **Qb6 (d4b6)**. Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1.top_p0001#3264` (ply 75) [structural distance $d=61$, QAP obj=29]: the master played **Qa3**; under μ our *Qd4* plays the role of that game’s *Qa3*; cross-type roles: our *Rd1* as their *Ph7*, our *Na5* as their *ph3*; transferred move **d4b6 (endorses this move)**. (c) Retrieved from `ccr1.top_p0003#97` (ply 93) [structural distance $d=55$, QAP obj=20]: the master played **Qe2**; under μ our *Qd4* plays the role of that game’s *Qe5*; cross-type roles: our *Rd1* as their *Pf7*, our *rf7* as their *ph2*; transferred move **d4b6 (endorses this move)**. (d) Retrieved from `ccr1.top_p0000#1185` (ply 95) [structural distance $d=59$, QAP obj=12]: the master played **Qb4**; under μ our *Qd4* plays the role of that game’s *Qa6*; cross-type roles: our *Na5* as their *Pb6*, our *Rd1* as their *pg2*; transferred move **d4b6 (endorses this move)**. Synthesis: 4 retrieved analogs transferred to **Qb6** (summed QAP objective 72, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



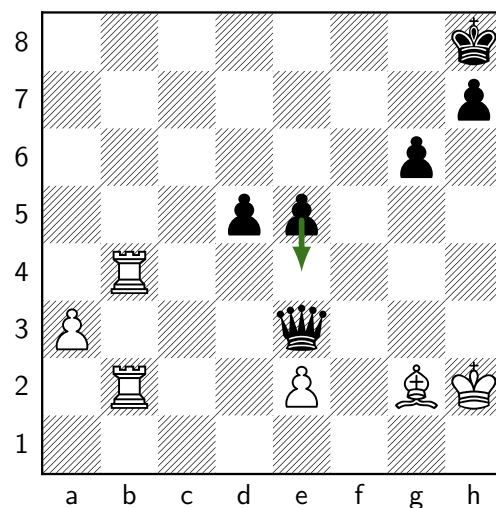
(a)



(b)

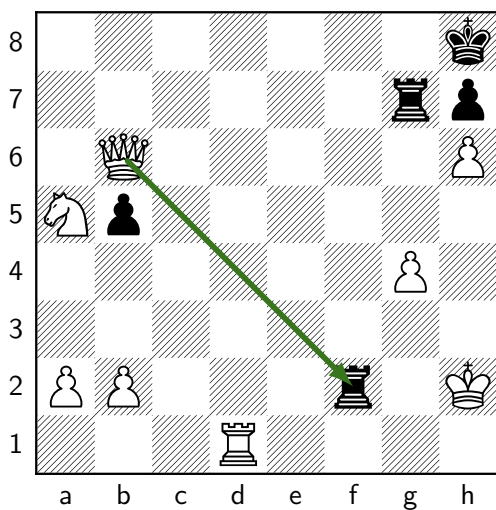


(c)

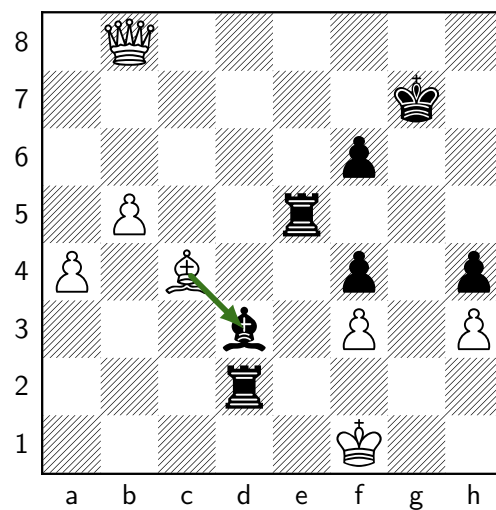


(d)

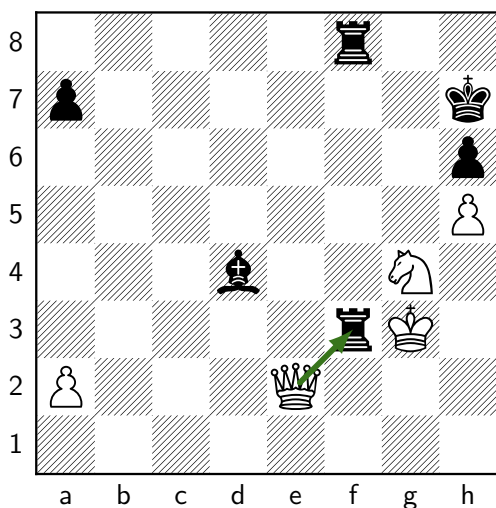
Figure 38: **Move 44, White: h6**. (a) Our position; Melanie plays h6 (h5h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1053` (ply 88) [structural distance $d=45$, QAP obj=15]: the master played Qa7+; under μ our Ph5 plays the role of that game’s Qa6 (a **cross-type** correspondence); cross-type roles: our Rd1 as their Ph2, our rf7 as their bd6; transferred move h5h6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1107` (ply 100) [structural distance $d=40$, QAP obj=34]: the master played Ra7+; under μ our Pa2 plays the role of that game’s Ra8 (a **cross-type** correspondence); cross-type roles: our Qb6 as their rb2, our Na5 as their rf2; transferred move a2a3. (d) Retrieved from `ccr1_top_p0001#852` (ply 73) [structural distance $d=38$, QAP obj=28]: the master played e4; under μ our Na5 plays the role of that game’s Pe4 (a **cross-type** correspondence); cross-type roles: our rg7 as their bg2; transferred move a5b7. Synthesis: 1 retrieved analog transferred to h6 (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



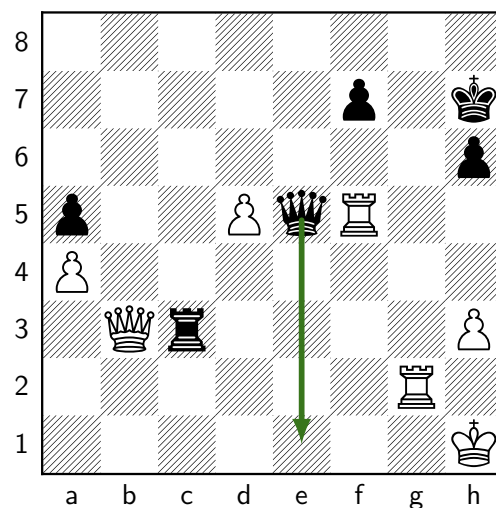
(a)



(b)

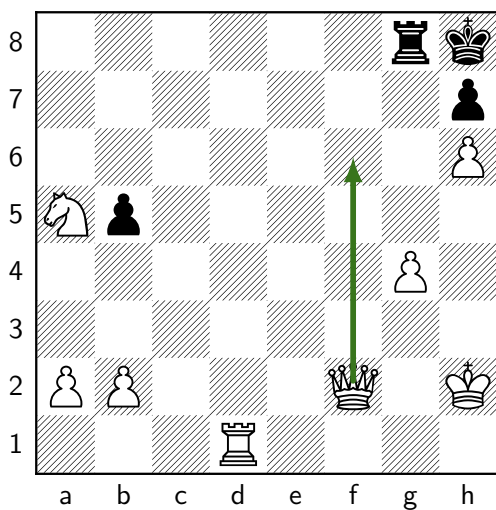


(c)

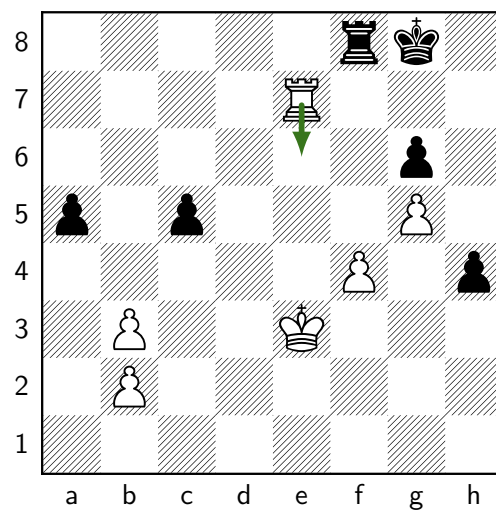


(d)

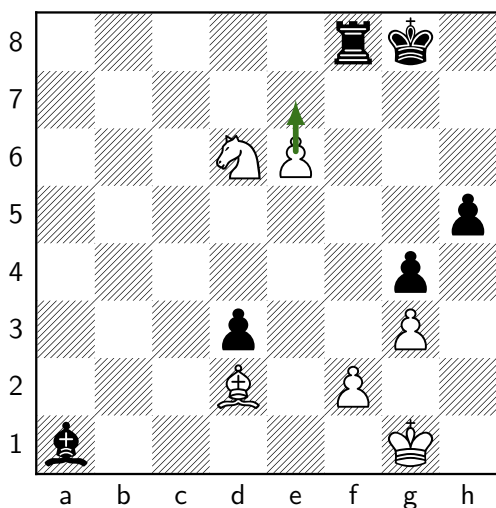
Figure 39: **Move 45, White: Qxf2**. (a) Our position; Melanie plays Qxf2 (b6f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2353` (ply 86) [structural distance $d=48$, QAP obj=67]: the master played Bxd3; under μ our Qb6 plays the role of that game's Bc4 (a **cross-type** correspondence); cross-type roles: our ph7 as their rd2, our rf2 as their bd3; transferred move b6f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#694` (ply 74) [structural distance $d=48$, QAP obj=35]: the master played Qxf3; under μ our Qb6 plays the role of that game's Qe2; cross-type roles: our Na5 as their bd4, our pb5 as their Ng4; transferred move b6f2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1974` (ply 87) [structural distance $d=46$, QAP obj=4]: the master played Qe1+; under μ our Qb6 plays the role of that game's Qe4; cross-type roles: our Rd1 as their qb3, our Na5 as their Rc3; transferred move b6f2 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Qxf2 (summed QAP objective 113, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



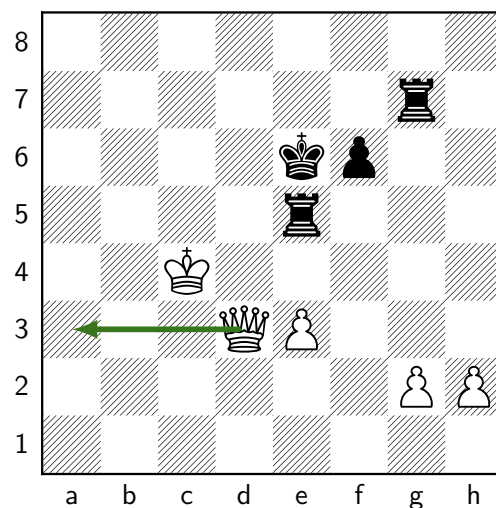
(a)



(b)

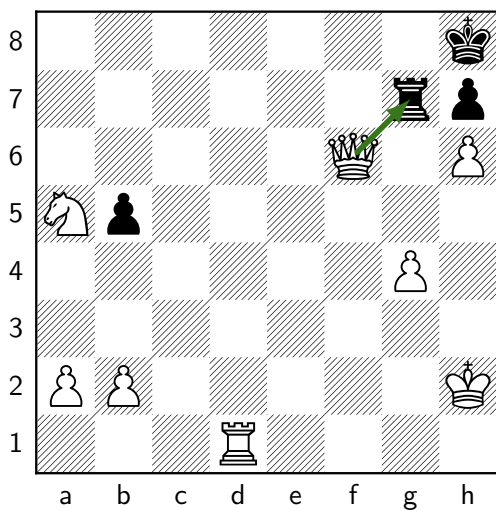


(c)

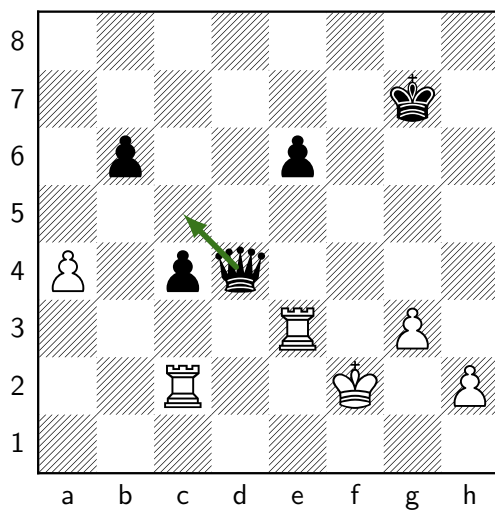


(d)

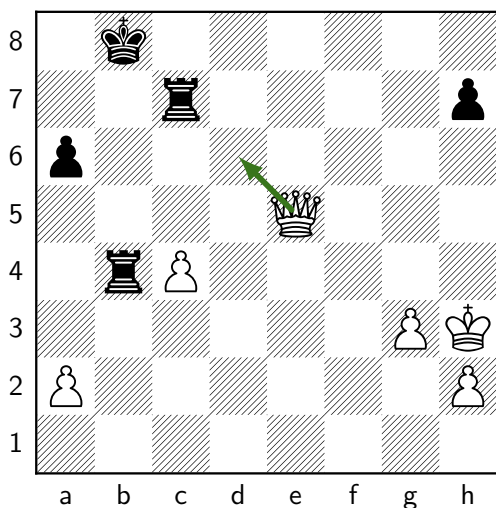
Figure 40: **Move 46, White: Qf6+ (f2f6).** (a) Our position; Melanie plays Qf6+ (f2f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2654` (ply 94) [structural distance $d=29$, QAP obj=25]: the master played Re6; under μ our Qf2 plays the role of that game's Re7 (a **cross-type** correspondence); cross-type roles: our Rd1 as their ph4, our Na5 as their pc5; transferred move f2c5. (c) Retrieved from `ccr1_top_p0000#86` (ply 96) [structural distance $d=29$, QAP obj=18]: the master played e7; under μ our Pa2 plays the role of that game's Pe6; cross-type roles: our Rd1 as their ba1, our pb5 as their Bd2; transferred move a2a3. (d) Retrieved from `ccr1_top_p0002#1754` (ply 100) [structural distance $d=30$, QAP obj=15]: the master played Qa3; under μ our Qf2 plays the role of that game's Qd3; cross-type roles: our Rd1 as their Ph2, our Na5 as their rg7; transferred move f2h4. Synthesis: no retrieved analog transferred to Qf6+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



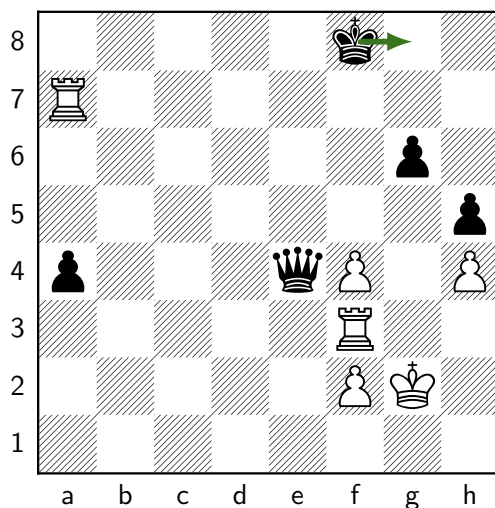
(a)



(b)



(c)



(d)

Figure 41: **Move 47, White: Qxg7#.** (a) Our position; Melanie plays Qxg7# (f6g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2473` (ply 101) [structural distance $d=23$, QAP obj=78]: the master played Qc5; under μ our Qf6 plays the role of that game’s Qd5; cross-type roles: our Na5 as their Pb6; transferred move f6d4. (c) Retrieved from `ccr1_top_p0000#3967` (ply 104) [structural distance $d=39$, QAP obj=57]: the master played Qd6; under μ our Qf6 plays the role of that game’s Qe5; cross-type roles: our Na5 as their rb4, our Rd1 as their pa6; transferred move f6e5. (d) Retrieved from `ccr1_top_p0001#2834` (ply 87) [structural distance $d=24$, QAP obj=46]: the master played Kg8; under μ our Kh2 plays the role of that game’s Kf1; cross-type roles: our Na5 as their Ph5; transferred move h2g2. Synthesis: no retrieved analog transferred to Qxg7# – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.